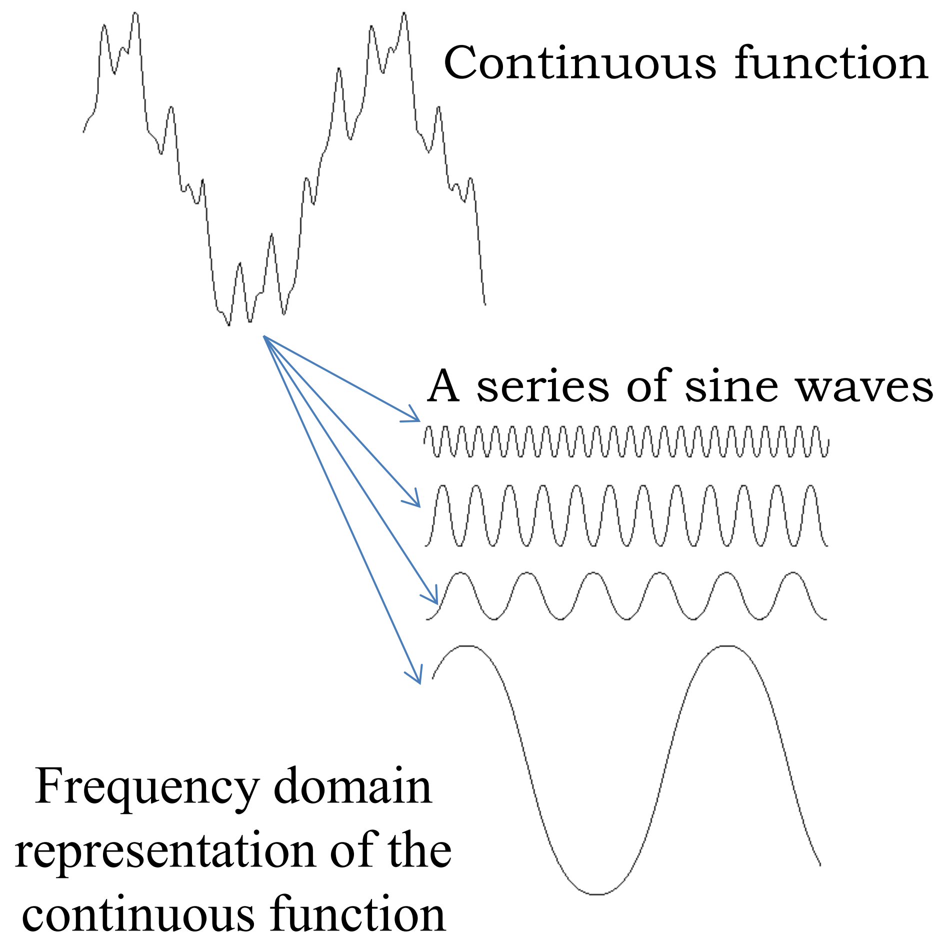
Image Enhancement in the Frequency Domain

Fourier Transform

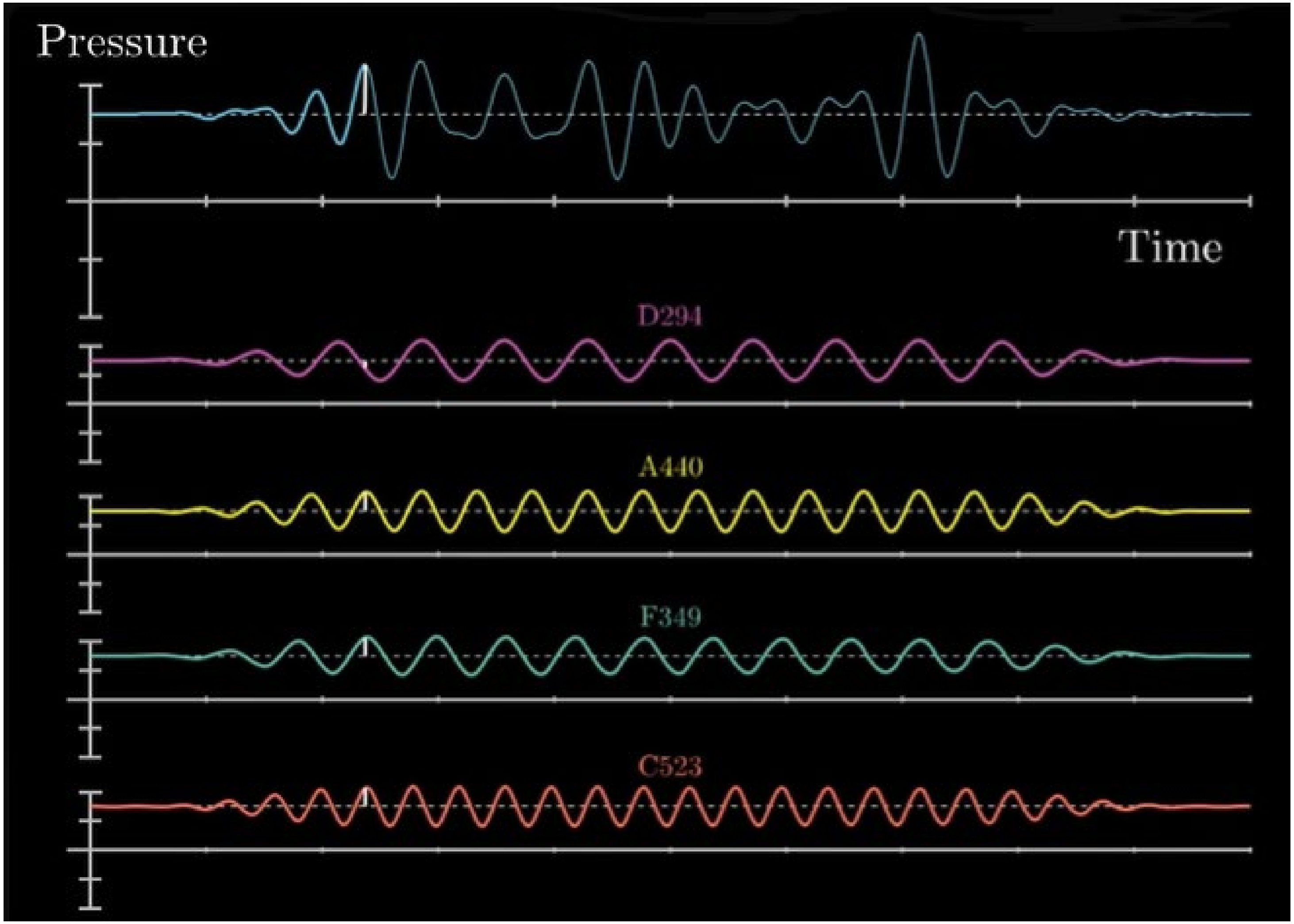
- A continuous function can be transformed into a series of sine waves. In Wikipedia, "A function of time (a signal) can be decomposed into its constituent frequencies" (https://en.wikipedia.org/wiki/Fourier_transform)

- This gives us a way to describe a function in terms of its frequencies



Function decomposition

For reference only



https://www.youtube.com/watch?v=spUNpyF58BY

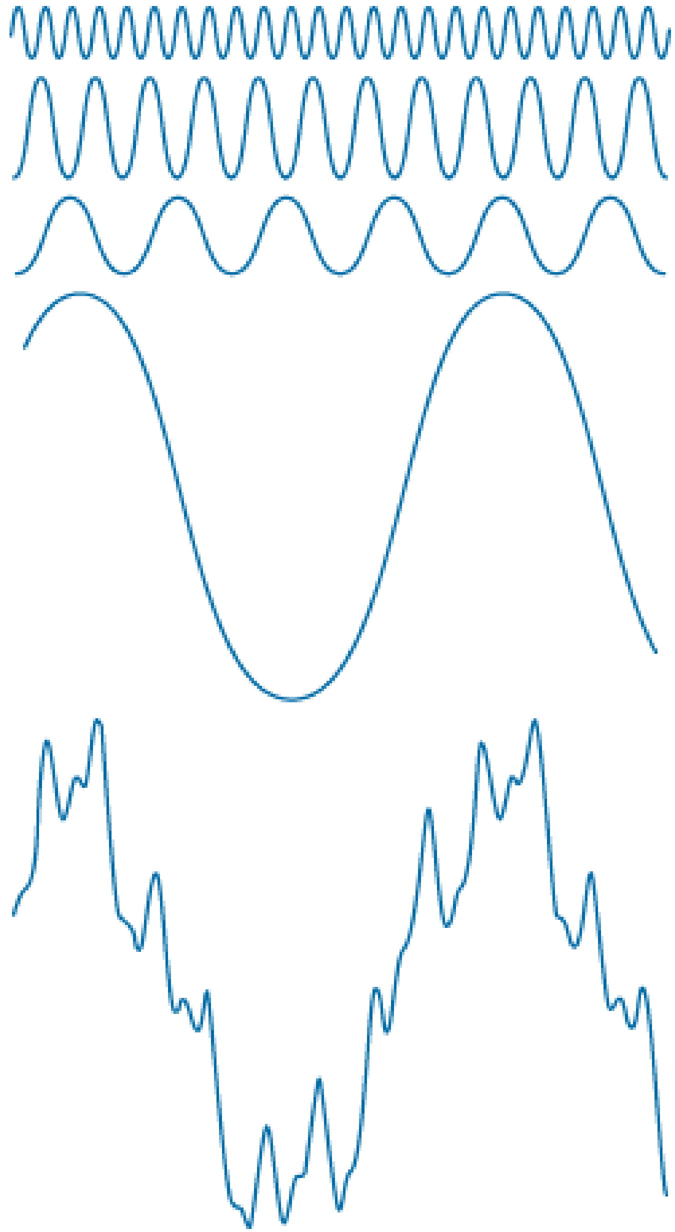
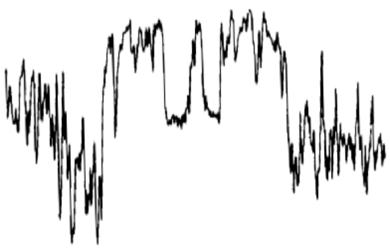


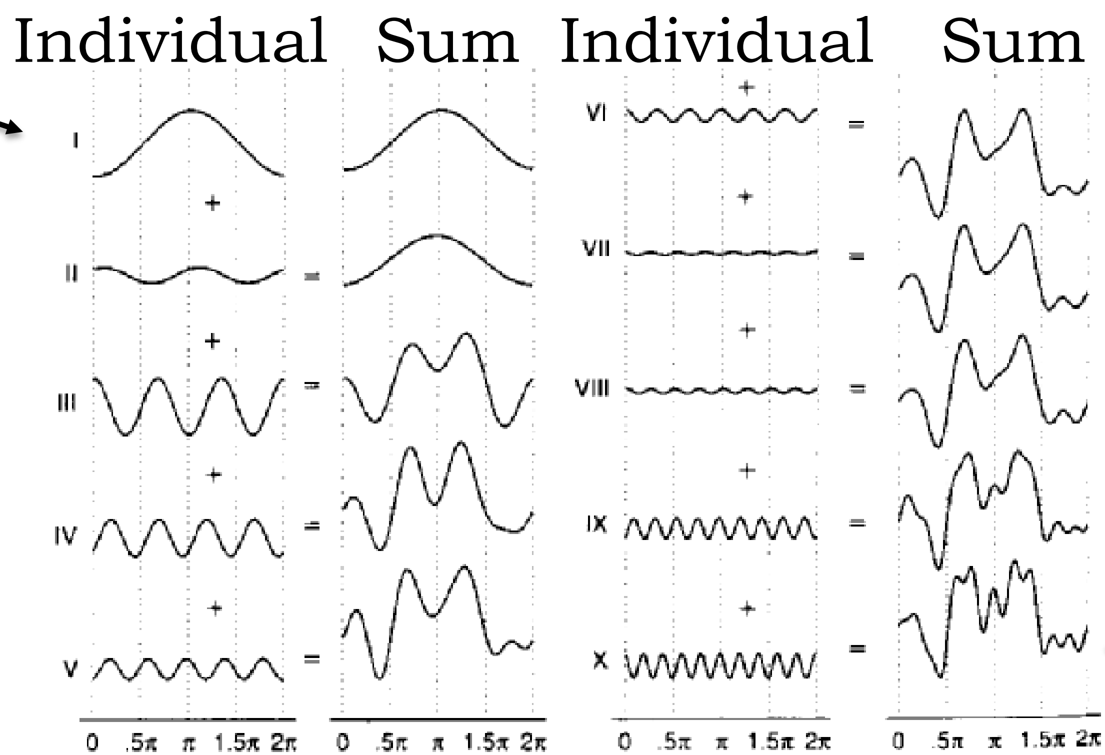
FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

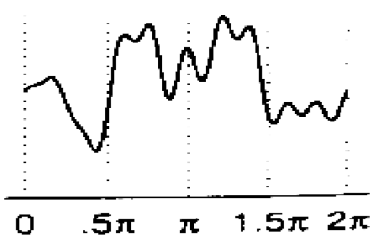
The function, f(x)
Input Signal

Sum of Frequencies



Input Signal





Output summing
first 10 frequencies from FT

Fourier Transform

http://en.wikipedia.org/wiki/Fourier_transform

- One-dimensional Fourier transform (based on the indefinite integral defined below) may have infinite frequency *u*. It is a weighted integral of sines and cosines.

$$F(u) = \int_{-\infty}^{\infty} f(x)[\cos 2\pi ux - i \sin 2\pi ux]dx$$

$$f(x) = \int_{-\infty}^{\infty} F(u)[\cos 2\pi ux + i \sin 2\pi ux]du$$

Inverse Fourier Transform

Sine wave: http://en.wikipedia.org/wiki/Sine_wave

Fourier Transform

http://en.wikipedia.org/wiki/Fourier_transform

- One-dimensional Fourier transform (based on the indefinite integral defined below) may have infinite frequency *u*.

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi ux} dx$$

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{i2\pi ux} du$$

Inverse Fourier Transform

Sine wave: http://en.wikipedia.org/wiki/Sine_wave
Euler's formula: https://en.wikipedia.org/wiki/Euler%27s_formula
The trigonometric functions are represented by using the complex exponential function

Discrete Fourier Transform (DFT)

The discrete Fourier transformation (definite summation) of the sampled f(x) is:

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{-j2\pi xu/N}$$

for u = 0, 1, 2, 3, . . . , N-1

Inverse:

$$f(x) = \sum_{u=0}^{N-1} F(u) e^{j2\pi xu/N}$$

Exponential function: http://en.wikipedia.org/wiki/Exponential_function

Sometimes this is written with the (1/N) switched.

$$F(u) = \sum_{x=0}^{N-1} f(x) e^{-j2\pi xu/N}$$

for u = 0, 1, 2, 3, . . . , N-1

$$f(x) = \boxed{\frac{1}{N}} \sum_{u=0}^{N-1} F(u) e^{j2\pi xu/N}$$

Exponential function: http://en.wikipedia.org/wiki/Exponential_function

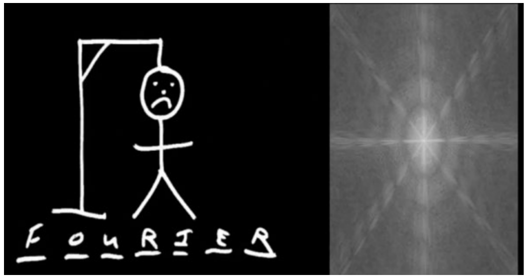
2D DFT

- In the two-variable case, the DFT pair is:

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$$

$$f(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M+vy/N)}$$

Live Fourier Transform Demo:
https://www.youtube.com/watch?v=qa1ZxK9Y1Tw

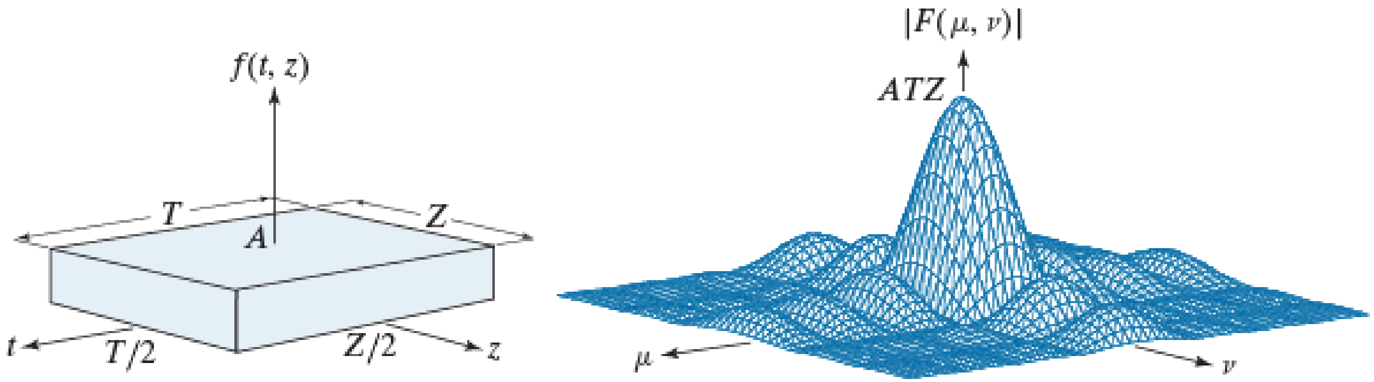


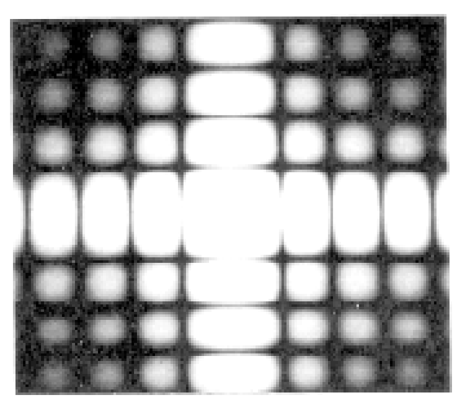
A Faster Fast Fourier Transform:
http://spectrum.ieee.org/computing/software/a-faster-fast-fourier-transform
https://en.wikipedia.org/wiki/Fast_Fourier_transform

Example

a b

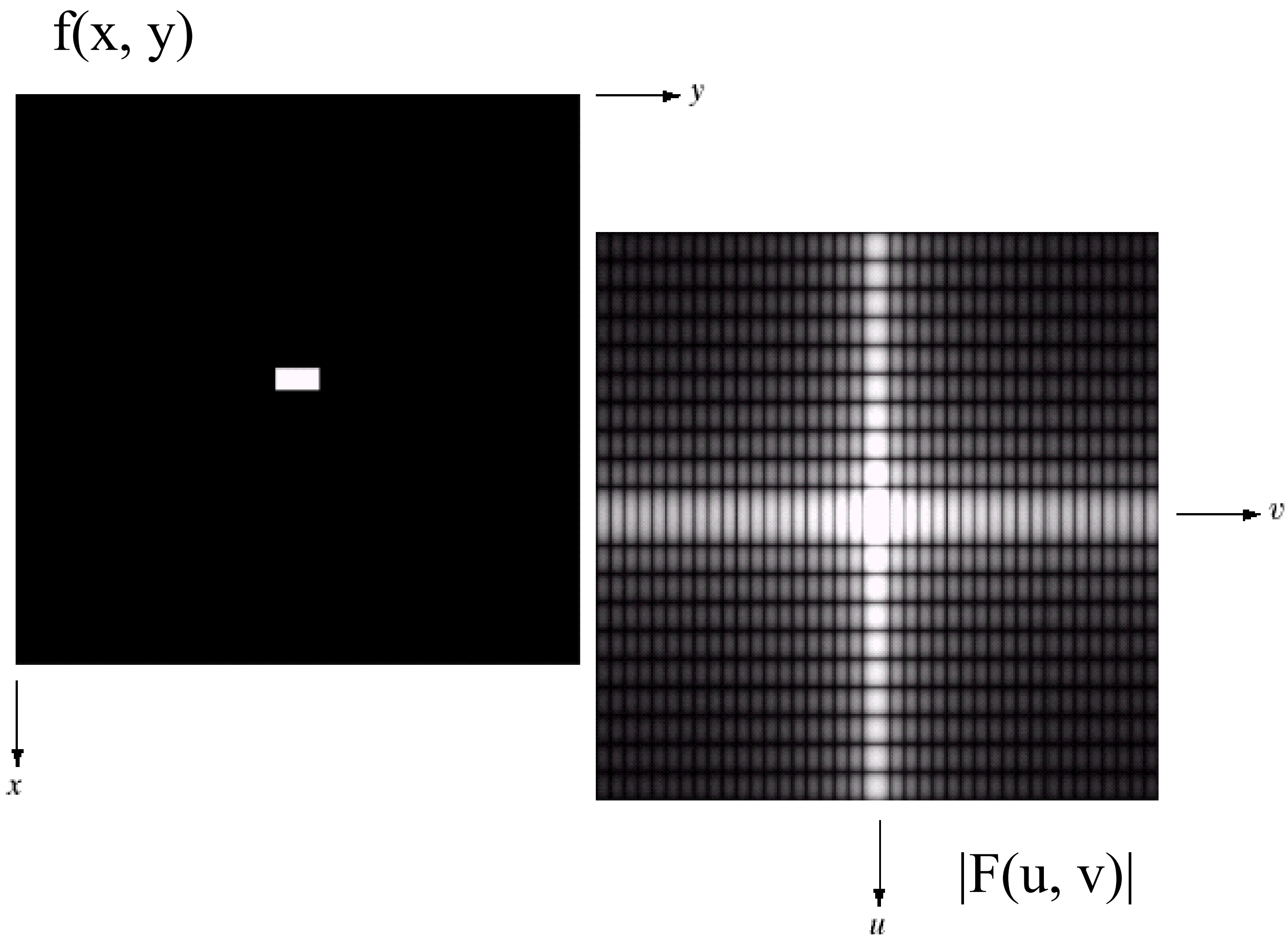
FIGURE 4.14
(a) A 2-D function and (b) a section of its spectrum. The box is longer along the t-axis, so the spectrum is more contracted along the μ-axis.





(c)

- We show |F(u,v)| and log(1 + |F(u,v)|)
- Note, these values have also been shifted such that F(0,0) is at F(N/2, N/2)



a b

FIGURE 4.3
(a) Image of a 20×40 white rectangle on a black background of size 512×512 pixels.
(b) Centered Fourier spectrum shown after application of the log transformation given in Eq. (3.2-2). Compare with Fig. 4.2.

Image Mean or Average Value

- Consider the mean image intensity

$$\bar{f}(x,y) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x,y)$$

- Consider F(0,0)

$$F(0,0) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x,y) e^{(0)}$$

Rotation

- Introduce polar coordinates,

  r = radius, theta θ/phi ϕ = angle

  $x = r \cos\theta \quad y = r \sin\theta \quad u = r \cos\phi \quad v = r \sin\phi$

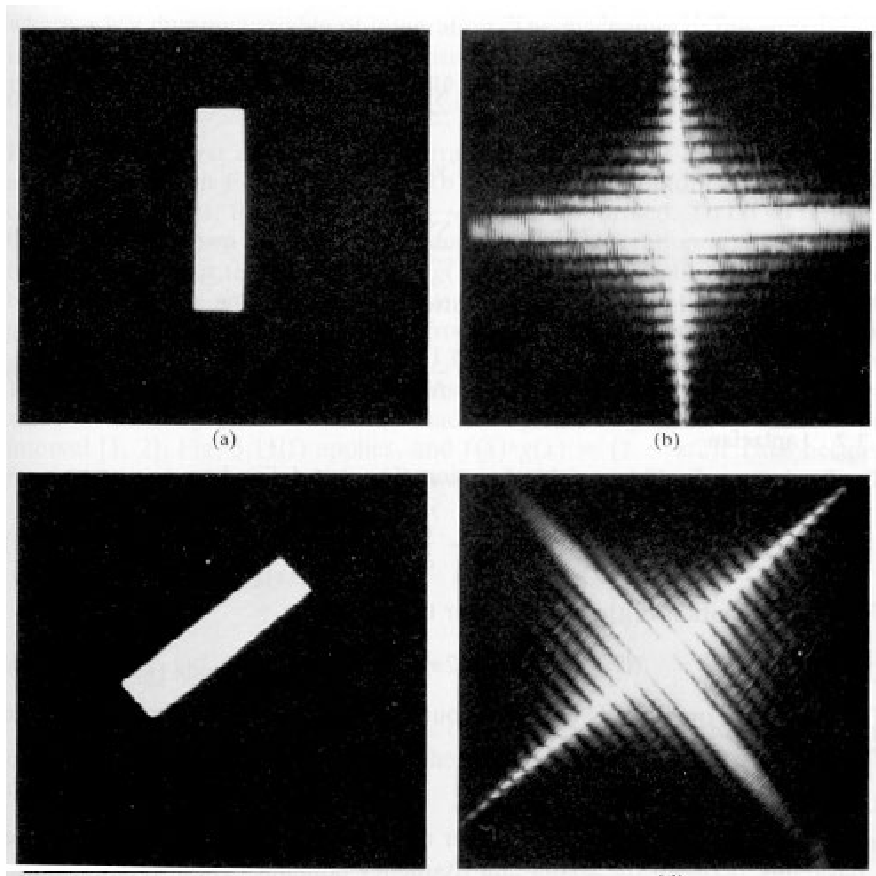
- Then, using the polar coordinates, we have

  $f(x,y) = f(r, \theta)$ and $F(u,v) = F(r, \phi)$

- Then $f(r, \theta + \theta_0) \Leftrightarrow F(r, \phi + \theta_0)$

Rotation: Example

$f(x,y) = f(r, \theta)$ and $F(u,v) = F(r, \phi)$



$f(x, y)$

$|F(u, v)|$

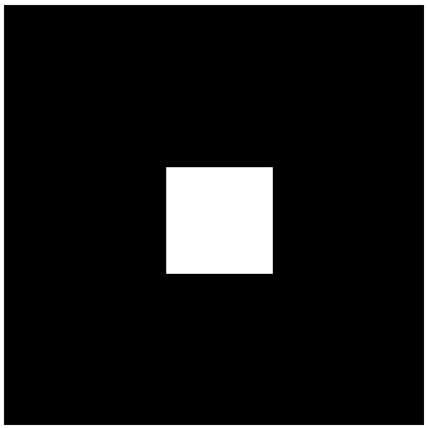
$f(r, \theta + \theta_0) \Leftrightarrow F(r, \phi + \theta_0)$

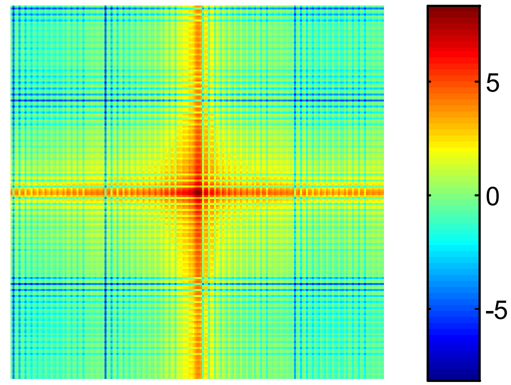
The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Translation properties of FT

$$f(x-x_0, y-y_0) \Leftrightarrow F(u,v)\, e^{-j2\pi(ux_0/M + vy_0/N)}$$

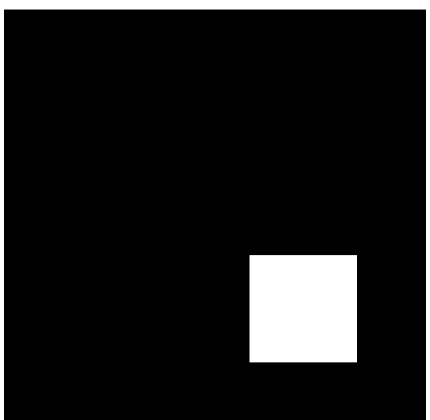
f(x, y)

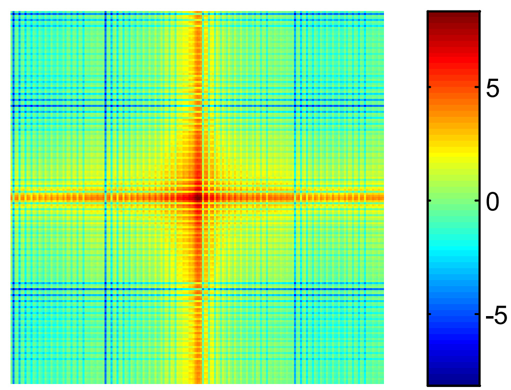




|F(u, v)|

f(x-x_0, y-y_0)





|F(u, v)|
Magnitude of the exponential function is one

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Filtering in
Frequency Domain

Filtering in Frequency Domain

- One image = two representations

$f(x,y)$

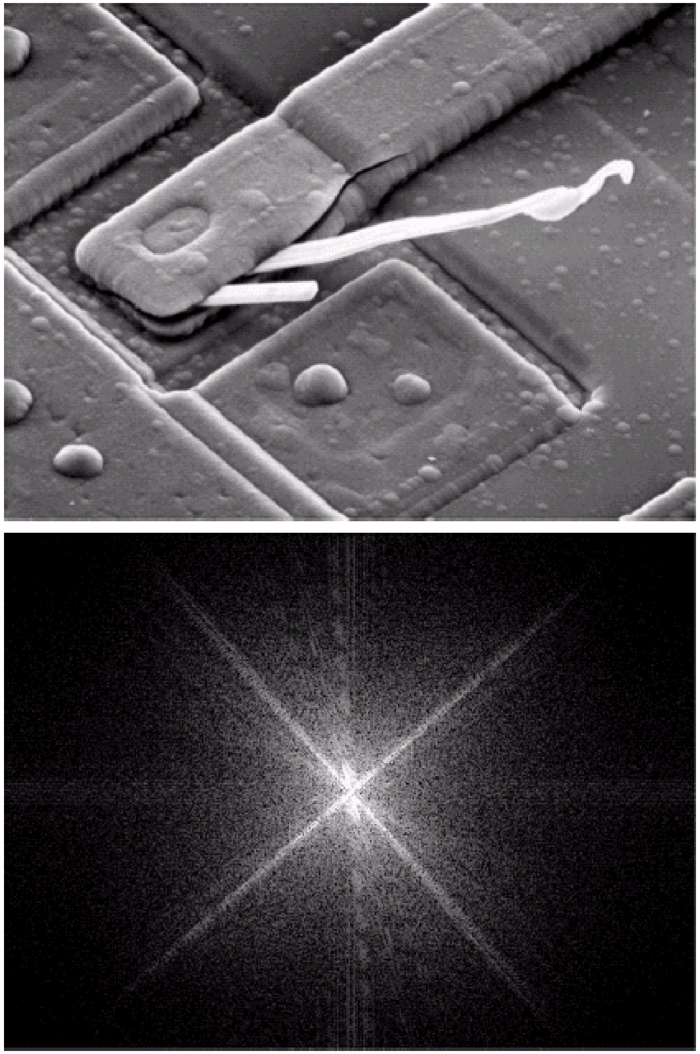
Spatial

(a collection of pixels)

$|F(u,v)|$

Frequency

(rate of change of intensity values or grey level)

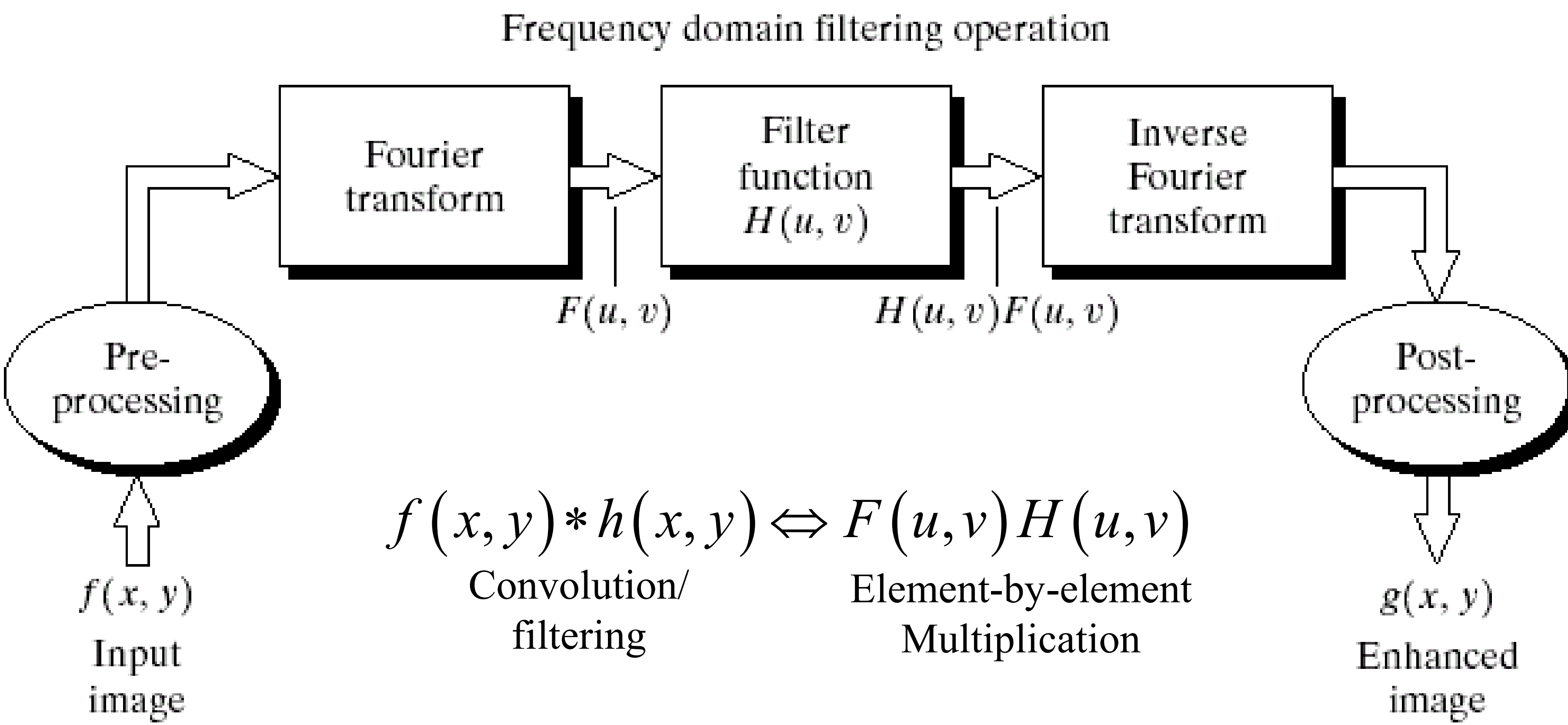


a
b

FIGURE 4.4 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

Strong edges at $\pm 45^{o}$ of the spectrum.

Filtering in Frequency Domain



$$f(x,y) * h(x,y) \Leftrightarrow F(u,v)H(u,v)$$

Convolution/ filtering

Element-by-element Multiplication

FIGURE 4.5 Basic steps for filtering in the frequency domain.

$$g(x,y) = f(x,y) * h(x,y)$$

$$G(u,v) = F(u,v)H(u,v)$$

$$g(x,y) = \mathfrak{I}^{-1}\{G(u,v)\}$$

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Filtering in Frequency Domain

- Filtering Frequencies
  - Low Pass, High Pass, . . .

- Filter will take the form
  - G(u,v) = H(u,v) F(u,v) *(element-by-element multiplication)*

    - F(u,v) is DFT of f(x,y), DFT = Discrete Fourier Transform
    - H(u,v) is a filter, and it attenuates or selects frequencies
    - G(u,v) is the filtered version of F(u,v)

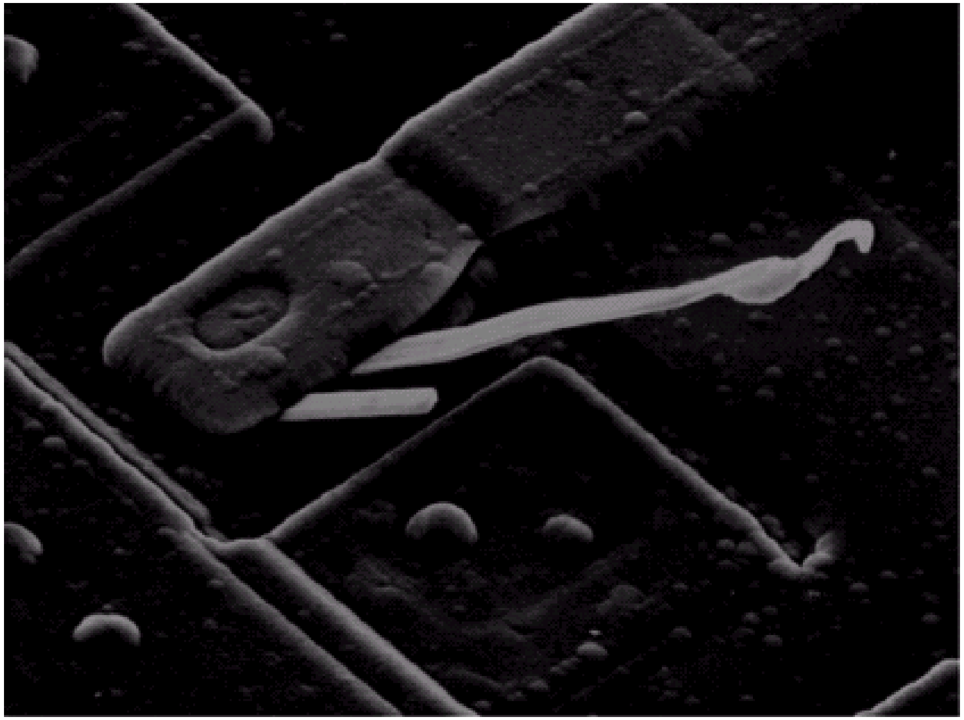
Example 1: Notch filter

$$H(u,v)=\begin{cases}0 & \text{if } (u,v)=(0,0)\\ 1 & \text{otherwise}\end{cases}$$

Note: The origin is located at
(M/2, N/2) for an MxN image matrix

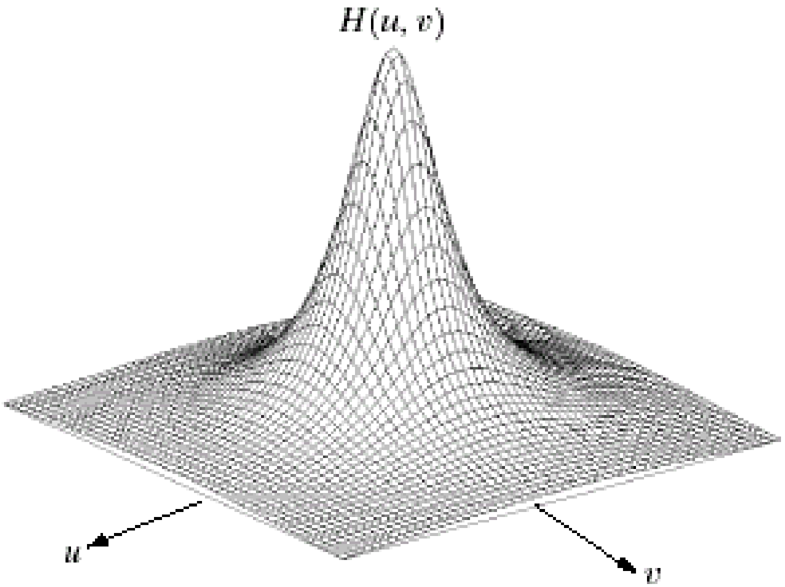
FIGURE 4.6 Result of filtering the image in Fig. 4.4(a) with a notch filter that set to 0 the $F(0,0)$ term in the Fourier transform.

Filtered image

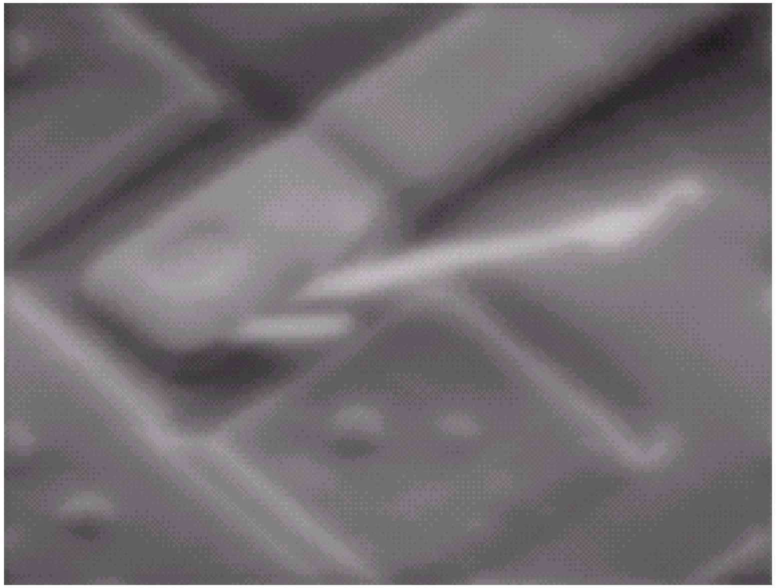


The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Example 2: Lowpass filter



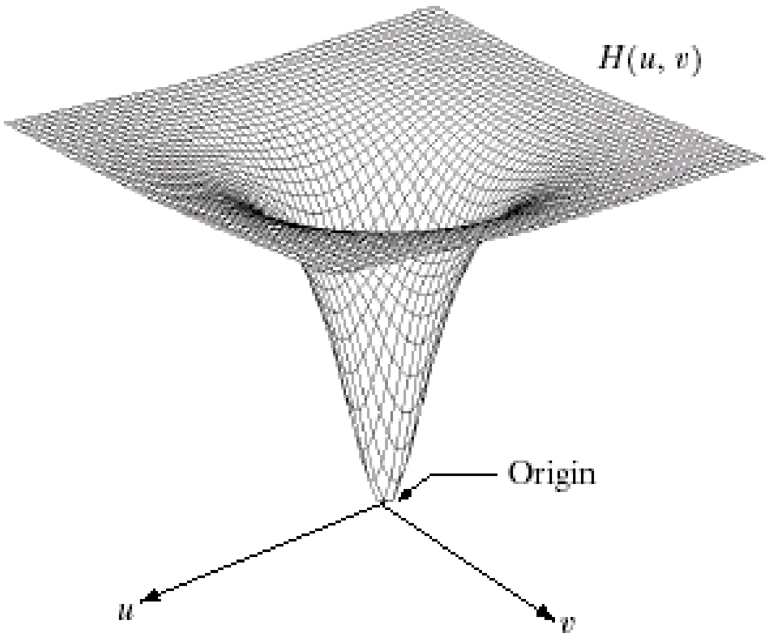
Filter
Circularly symmetric



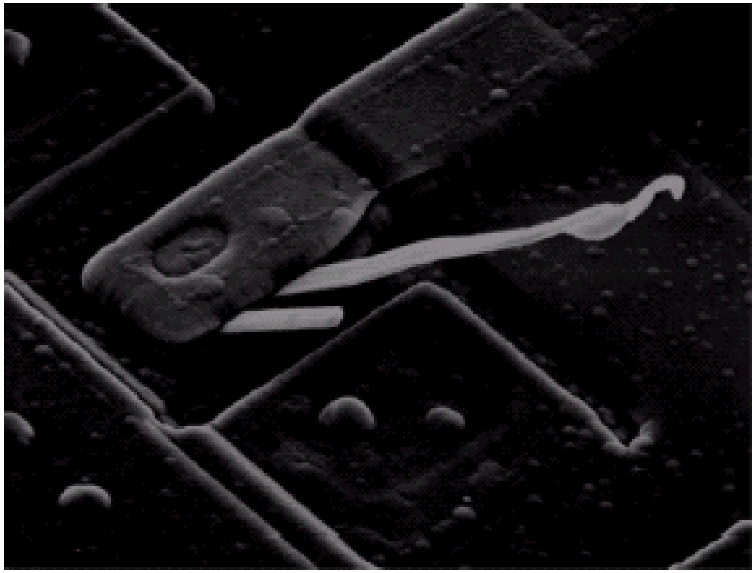
Filtered image

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Example 3: Highpass filter



Filter
Circularly symmetric



Filtered image

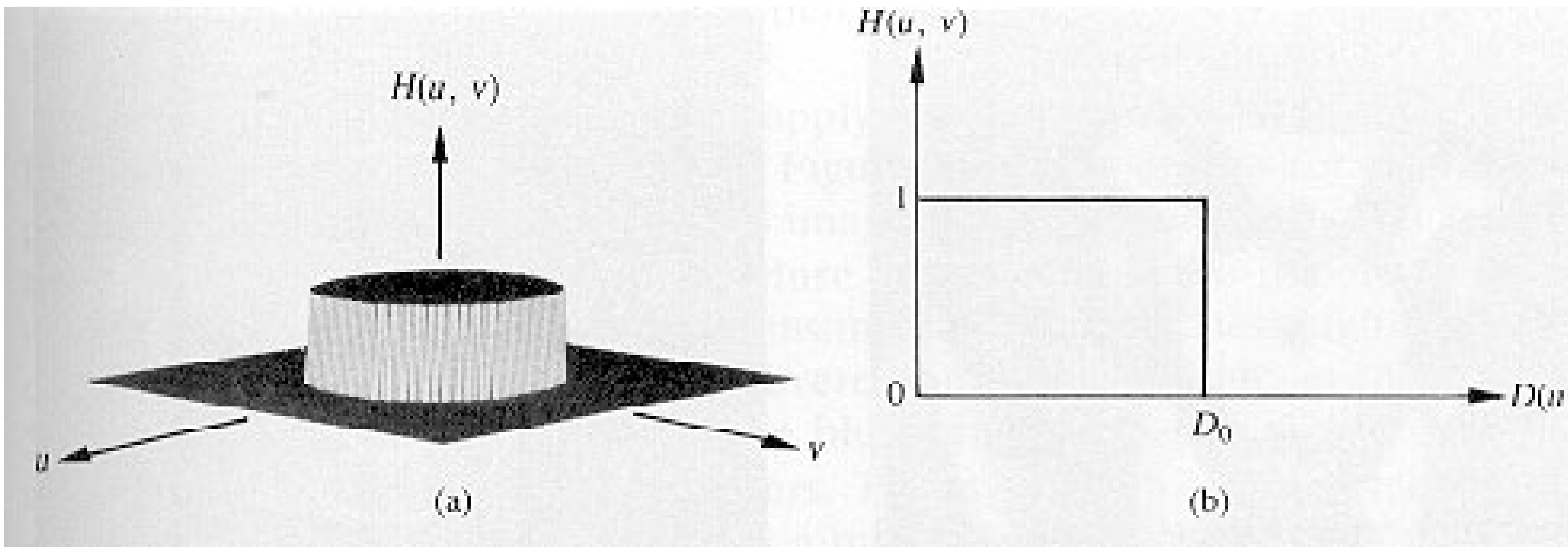
The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Smoothing Frequency-Domain Filters

Ideal Lowpass Filter (ILPF)

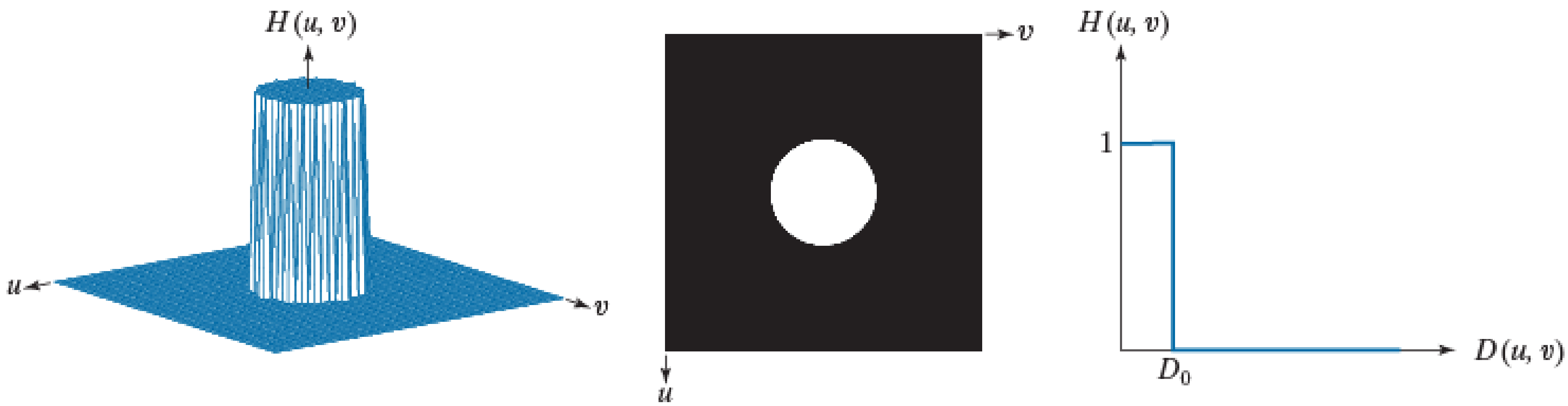
- Ideal Lowpass Filter
  - $$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) <= D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$
  - $D(u,v)$ is the distance from point (u,v) to the origin of the frequency plane; $D(u,v) = (u^2 + v^2)^{1/2}$
  - D_0 is a non-negative quantity

Graphical Representation of the Ideal Lowpass Filter



The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Graphical Representation of the Ideal Lowpass Filter



a b c

FIGURE 4.39 (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Function displayed as an image. (c) Radial cross section.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Image Power

- We can compute the amount of power a filter "encloses" from the total Power P_T

$$P_T = \sum_u \sum_v P(u,v)$$

$$P(u,v) = R(u,v)^2 + I(u,v)^2$$

(**R**eal and **I**maginary Components of DFT)

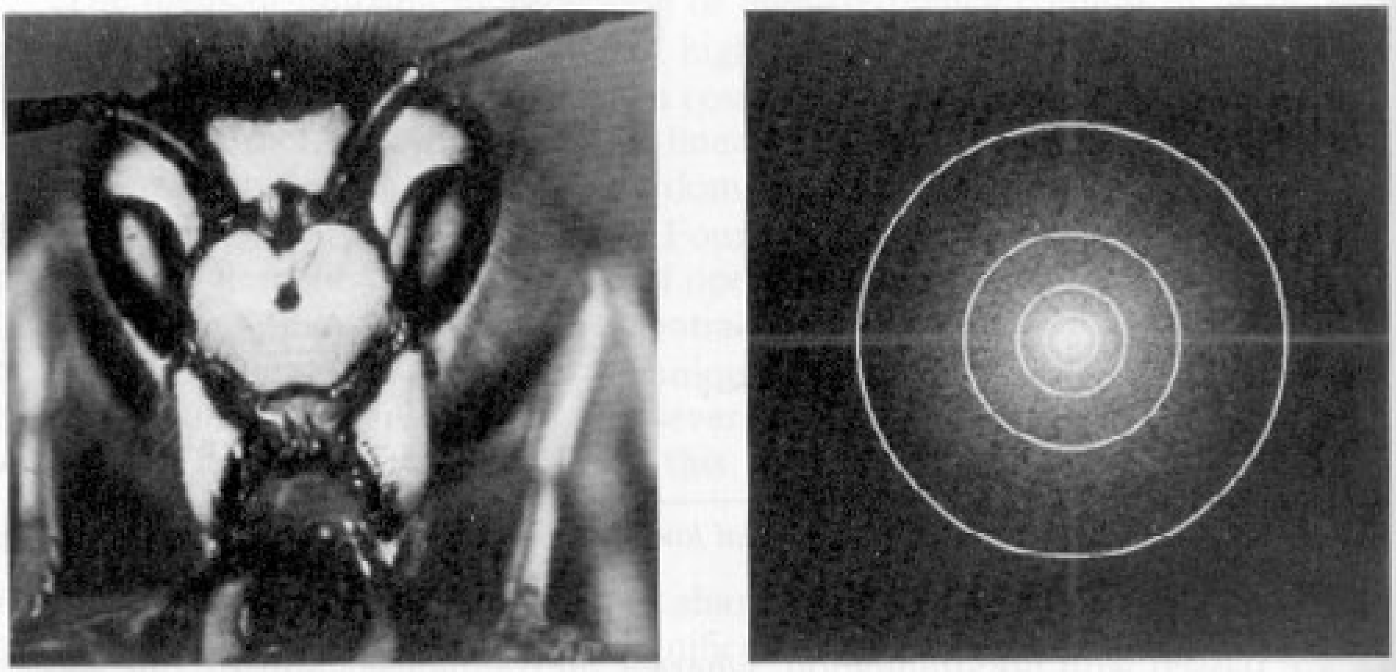
Image Power

- A filter, centered at the original of the frequency plane with radius *r*, enclosed β percent of power

$$\beta = \sum_{u} \sum_{v} \frac{P(u,v)}{P_T} \times 100\%$$

- where (u,v) are summed over the coordinates that lie within the circle or on its boundary

Example


(a) (b)

- A 512x512 image
- Filters with radii = 8,18,43,78,153 pixels
- Power = 90%, 93%, 95%, 99%, 99.5% respectively

A Faster Fast Fourier Transform: "Natural signals often have relatively few frequency components of significance."
http://spectrum.ieee.org/computing/software/a-faster-fast-fourier-transform

Filtering with Ideal filter

Original

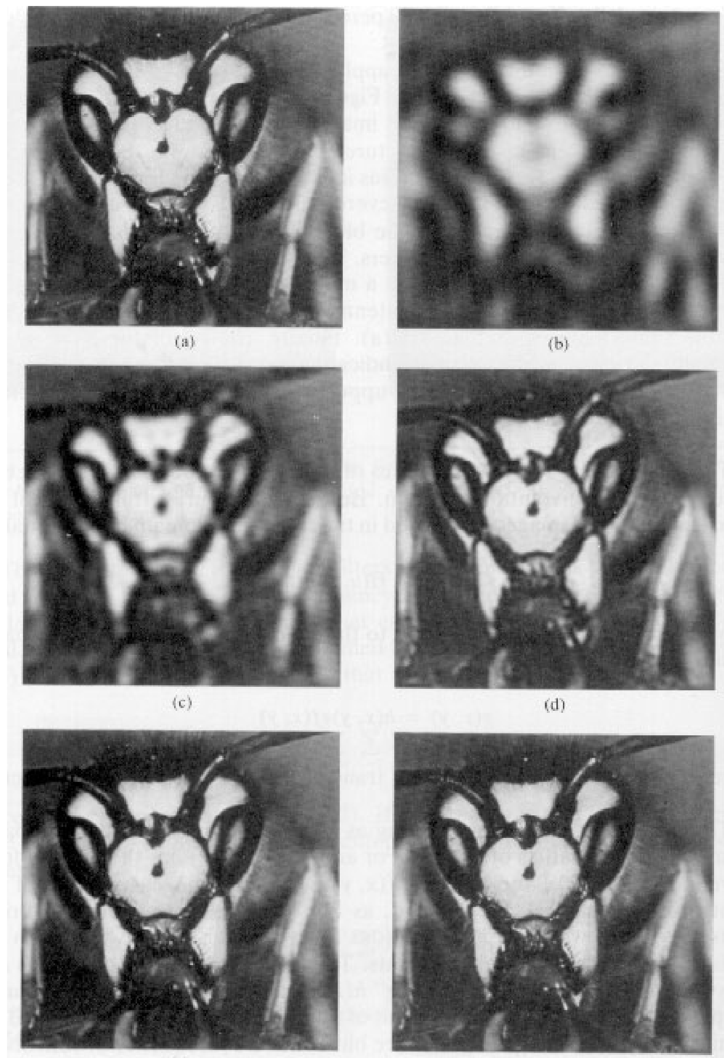
8 (Filter Radius)

18

43

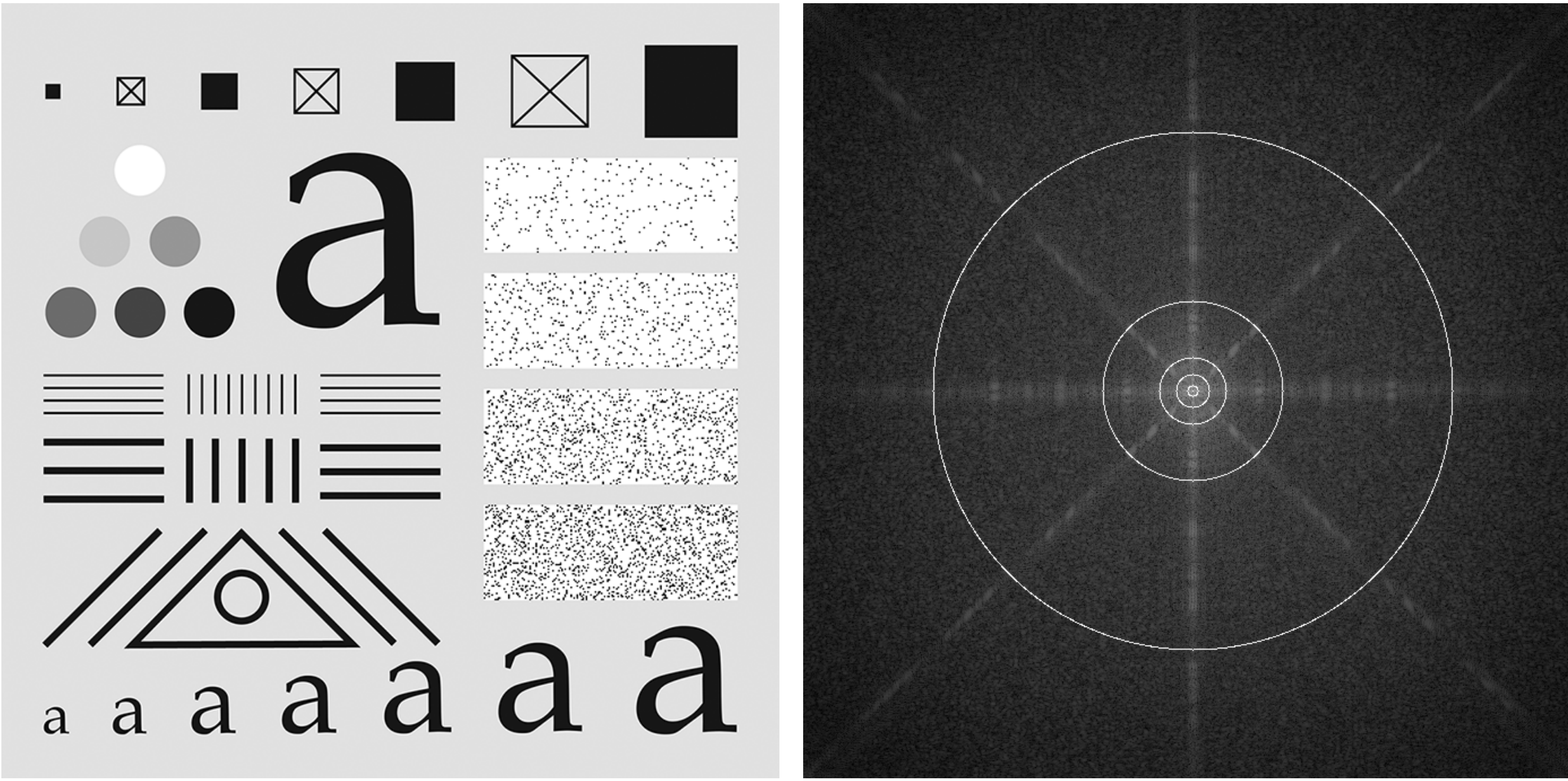
78

153



The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Another example

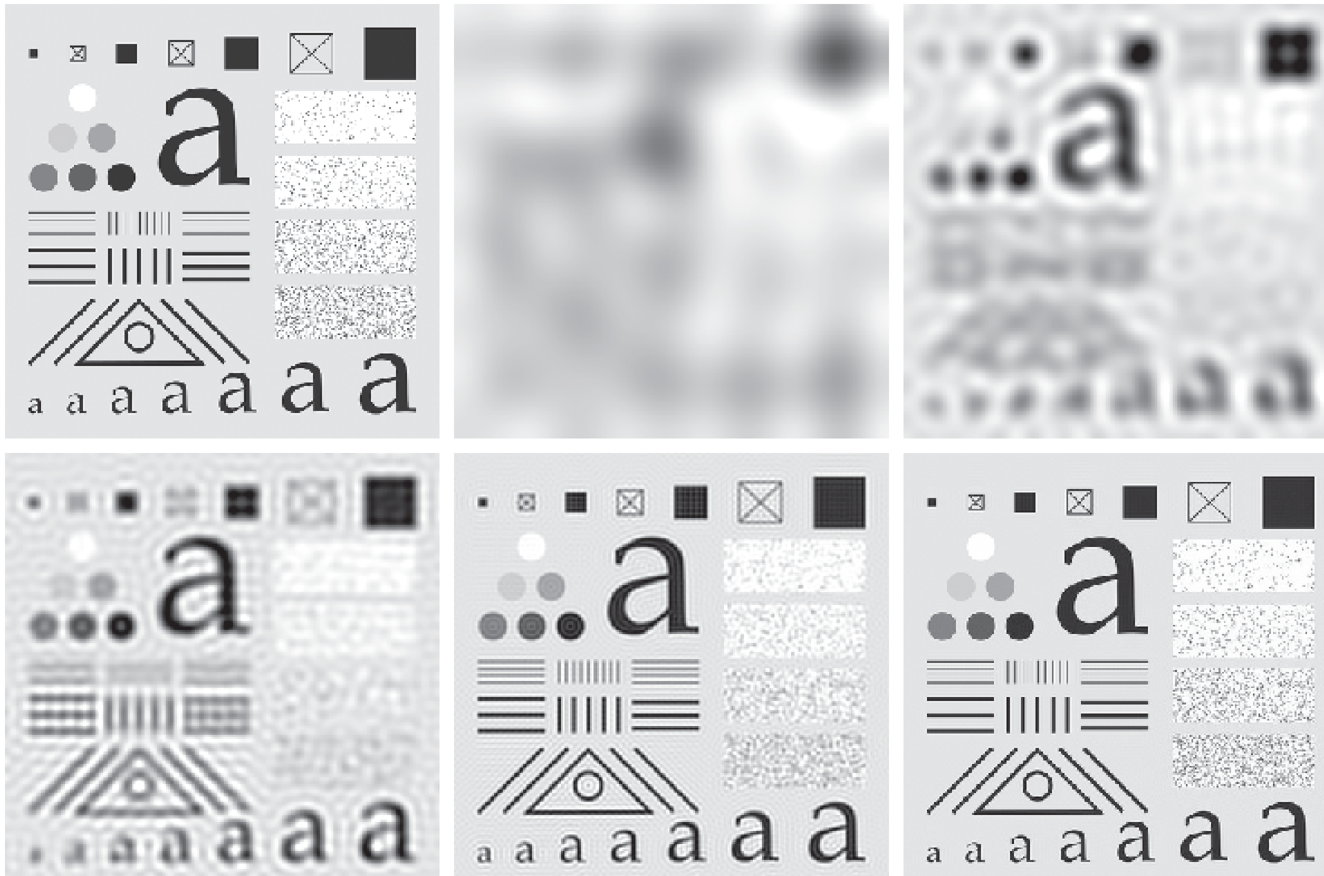


a b

FIGURE 4.40 (a) Test pattern of size 688 × 688 pixels, and (b) its spectrum. The spectrum is double the image size as a result of padding, but is shown half size to fit. The circles have radii of 10, 30, 60, 160, and 460 pixels with respect to the full-size spectrum. The radii enclose 86.9, 92.8, 95.1, 97.6, and 99.4% of the padded image power, respectively.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Filtering with Ideal filter



a b c
d e f

FIGURE 4.41 (a) Original image of size 688 × 688 pixels. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.40(b). The power removed by these filters was 13.1, 7.2, 4.9, 2.4, and 0.6% of the total, respectively. We used mirror padding to avoid the black borders characteristic of zero padding, as illustrated in Fig. 4.31(c).

Most of the sharp detail information in the image is contained in the 13% of the power removed by the filter.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Ideal Filter

- The sharp cut-off of the ideal filter in the frequency domain can result in ripples appearing in the image. It is called ringing artifact.

- We can design a filter with a smooth cut off.

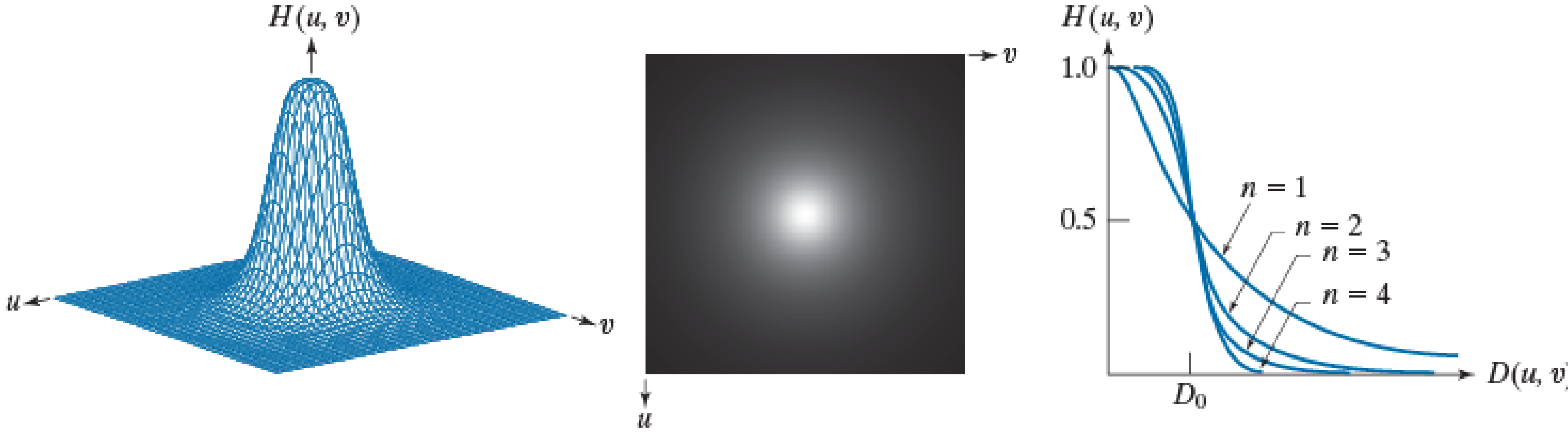
http://en.wikipedia.org/wiki/Ringing_artifacts

Butterworth Lowpass Filter (BLPF)

- This filter does not have a sharp discontinuity
  - Instead, it has a smooth transition
- A Butterworth filter of order **n** and cutoff frequency locus at a distance D_0 has the form

$$H(u,v) = \frac{1}{1 + [D(u,v)/D_0]^{2n}}$$

  - where D(u,v) is the distance from the center of the frequency plane, and D_0 is a positive quantity



a b c

FIGURE 4.45 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Function displayed as an image. (c) Radial cross sections of BLPFs of orders 1 through 4.

1. The BLPF transfer function does not have a sharp discontinuity that sets up a clear cutoff between passed and filtered frequencies.
2. No ringing artefact visible when n = 1. Very little artefact appears when n <= 2 (hardly visible).

Example (BLPFs) Order = 2.25

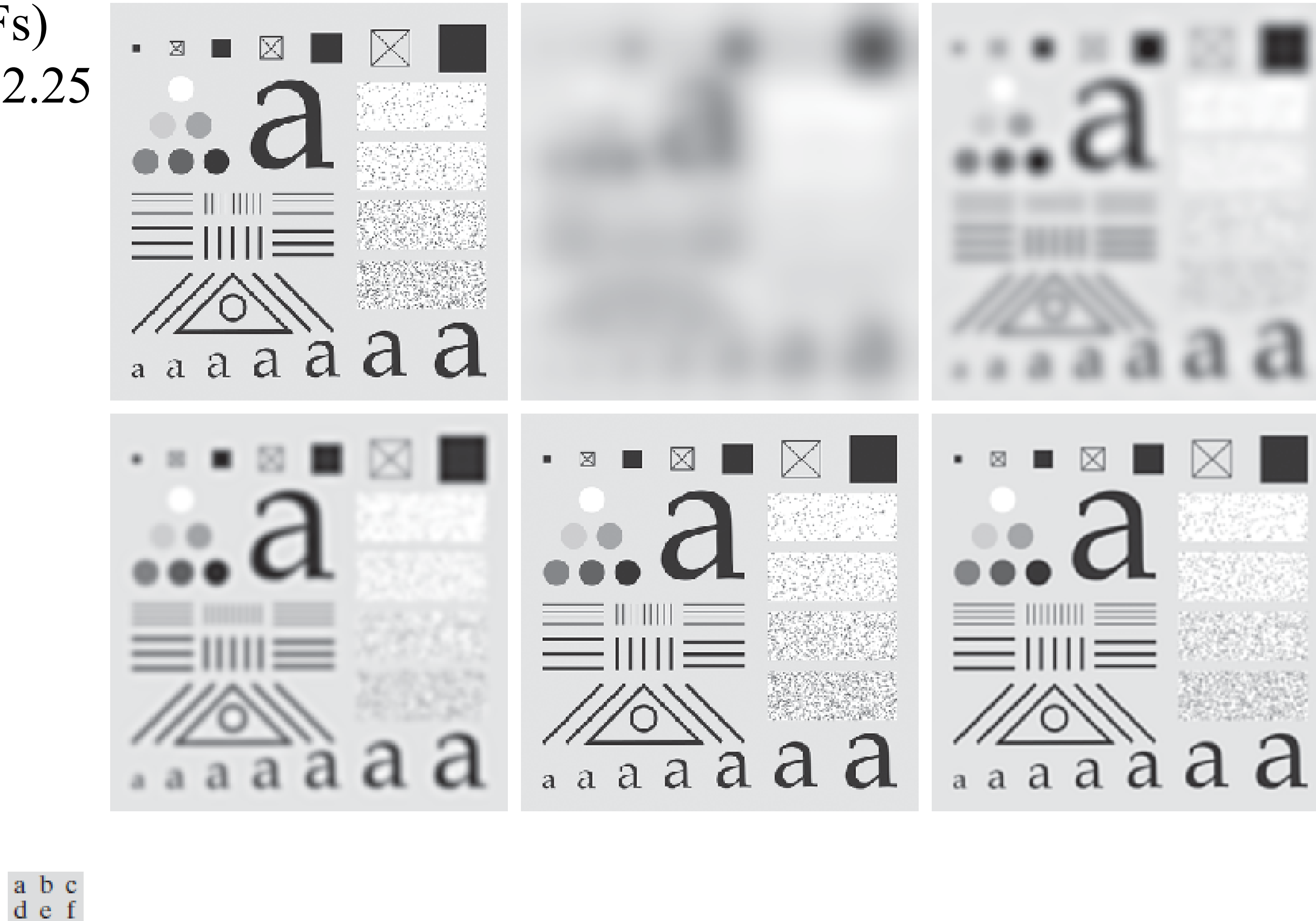


FIGURE 4.46 (a) Original image of size 688 × 688 pixels. (b)–(f) Results of filtering using BLPFs with cutoff frequencies at the radii shown in Fig. 4.40 and $n = 2.25$. Compare with Figs. 4.41 and 4.44. We used mirror padding to avoid the black borders characteristic of zero padding.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Slight Variation

- For these filters
  - We often want to define a cutoff frequency
  - For which H(u,v) is decreased to a certain fraction of its maximum value
  - For example: Let H(u,v) = 0.5 when D(u,v) = D_0 and α is 1

$$H(u,v) = \frac{1}{1 + \boxed{(\alpha)}[D(u,v)/D_0]^{2n}}$$

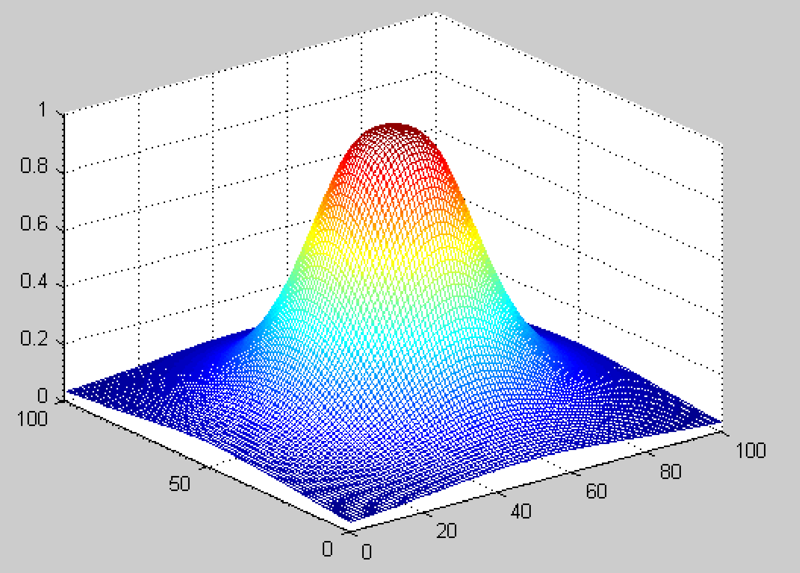
Slight Variation

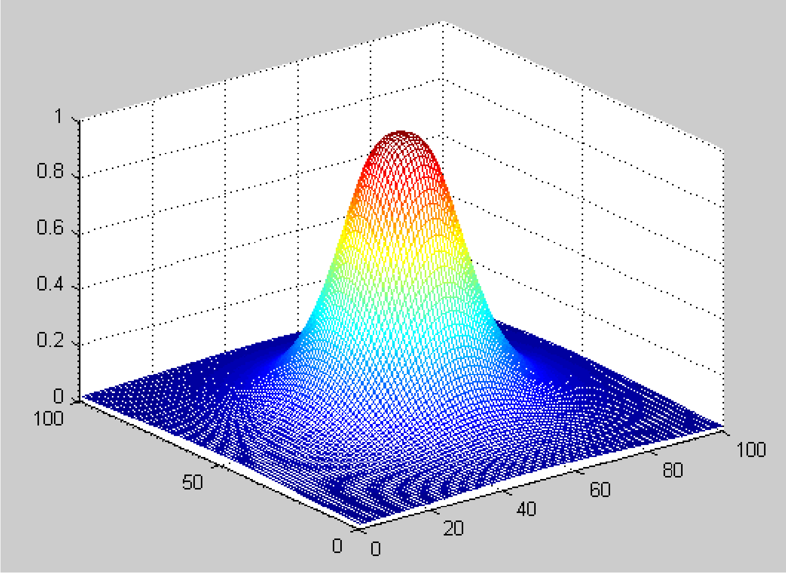
- Another example is to have H(u,v) be $1/\sqrt{2}$ (=0.707) of the maximum value of H(u,v) when D(u,v) = D_0 and α is $\sqrt{2}-1$ (=0.414).

$$H(u,v) = \frac{1}{1+(\sqrt{2}-1)[D(u,v)/D_0]^{2n}}$$

$$H(u,v) = \frac{1}{1+0.414[D(u,v)/D_0]^{2n}}$$

Examples

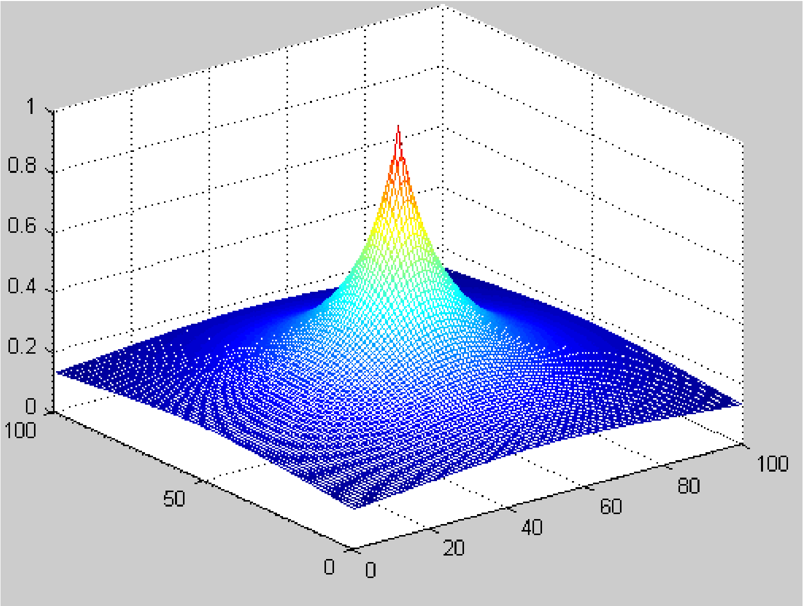


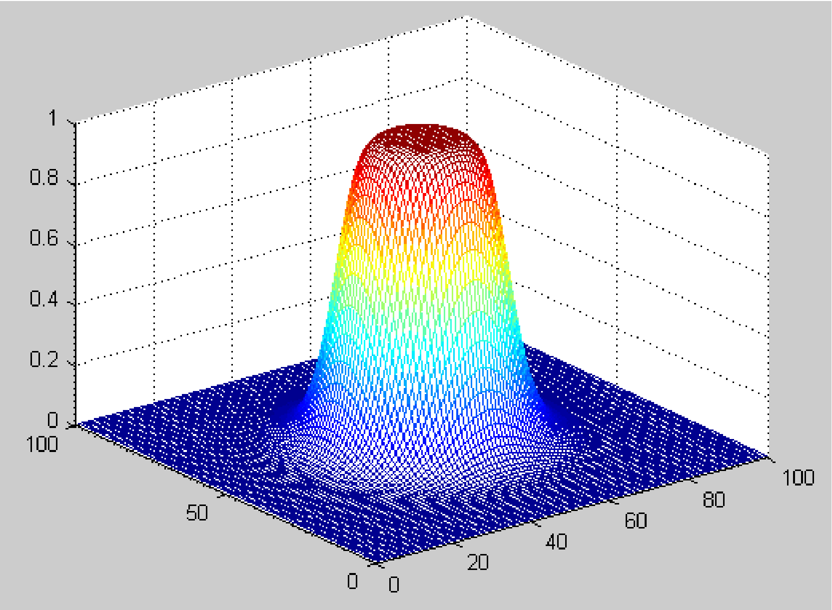


100 x 100
$D_0 = 20$
$\alpha = 0.414$
n = 3

100 x 100
$D_0 = 20$
$\alpha = 1$
n = 3

Examples





100 x 100
D_0 = 20
α = 1
n = 1

100 x 100
D_0 = 20
α = 1
n = 7

Gaussian Lowpass Filter (GLPF)

- This filter is one of the commonly used filters.

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

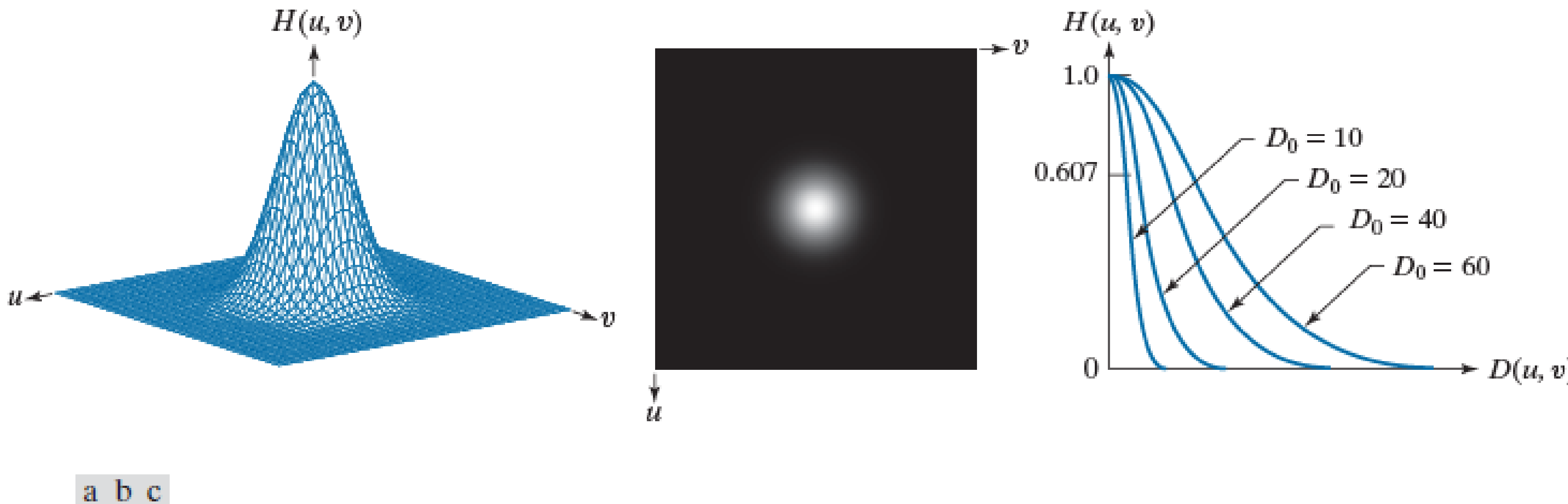
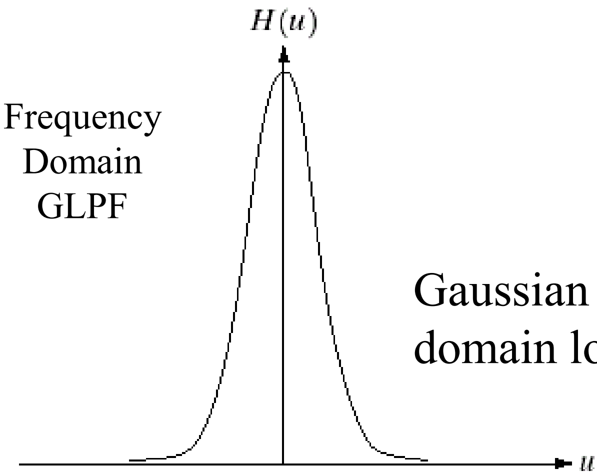


FIGURE 4.43 (a) Perspective plot of a GLPF transfer function. (b) Function displayed as an image. (c) Radial cross sections for various values of D_0.

Gaussian Lowpass Filter (GLPF)

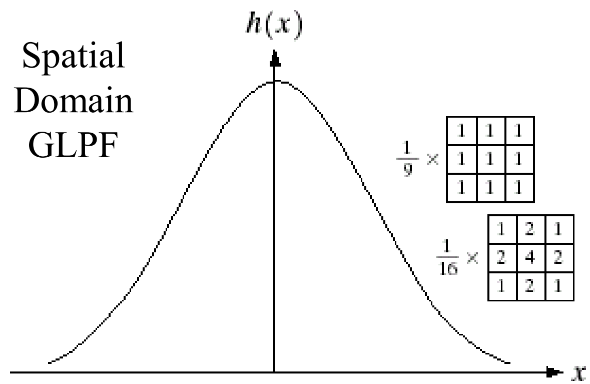
$$H(u,v) = e^{-D^2(u,v)/2D_0^2} \Leftrightarrow h(x,y) = \sqrt{2\pi}D_0 e^{-2\pi^2 D_0^2 D^2(x,y)}$$

Frequency
Domain
GLPF



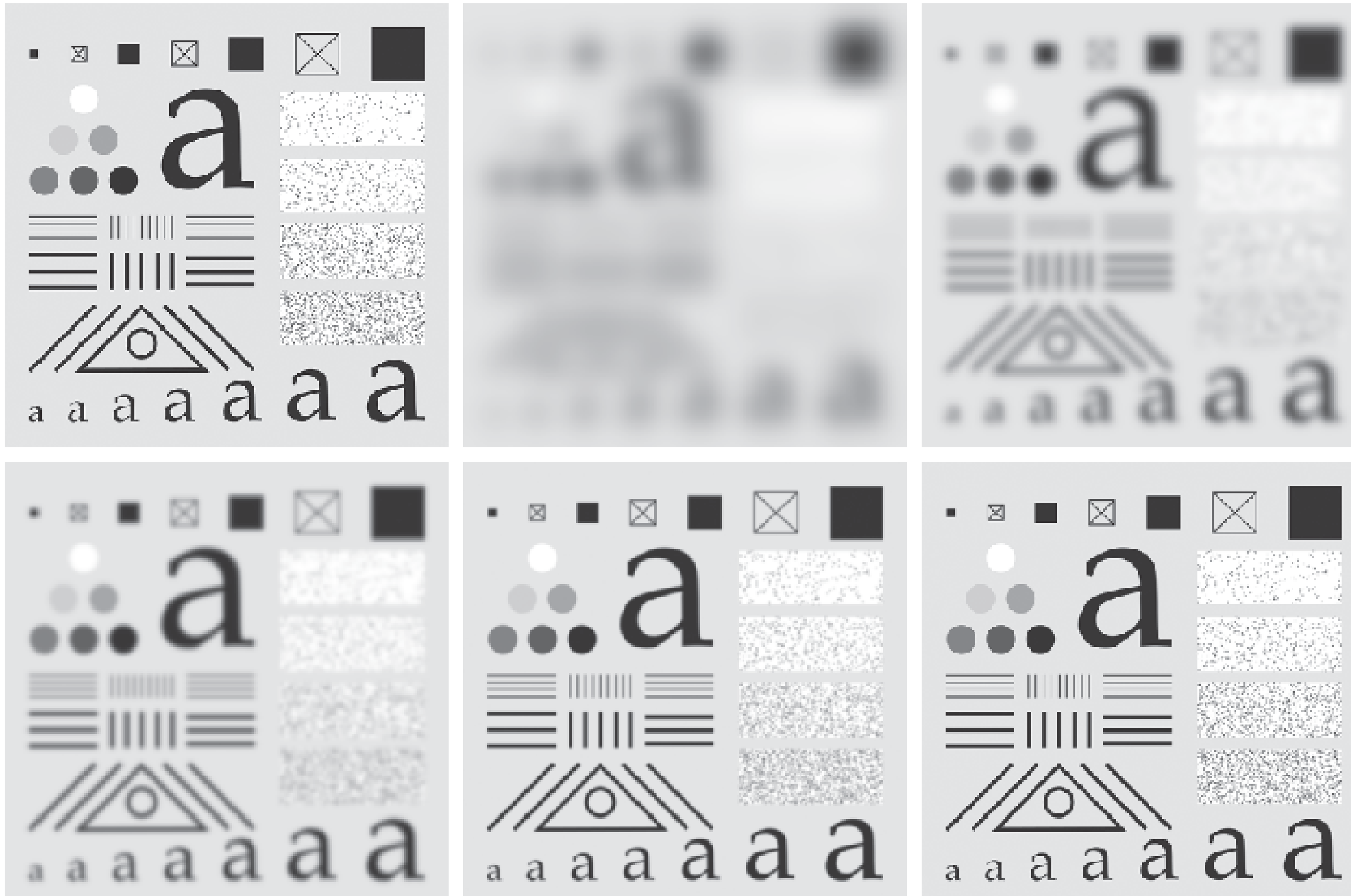
Gaussian frequency
domain lowpass filter

Spatial
Domain
GLPF



Corresponding spatial
domain lowpass filter

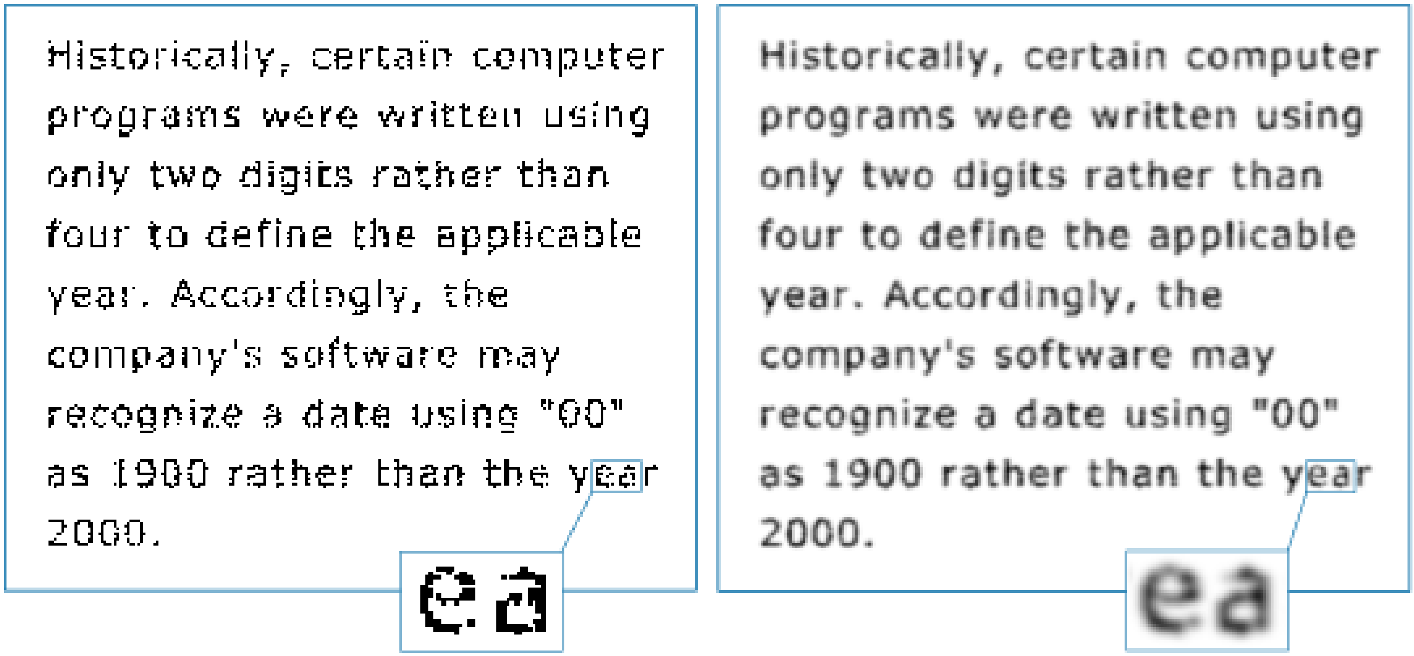
Example (GLPFs)



a b c
d e f

FIGURE 4.44 (a) Original image of size 688 × 688 pixels. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.40. Compare with Fig. 4.41. We used mirror padding to avoid the black borders characteristic of zero padding.

Example (GLPFs)



a b

FIGURE 4.48
(a) Sample text of low resolution (note the broken characters in the magnified view). (b) Result of filtering with a GLPF, showing that gaps in the broken characters were joined.

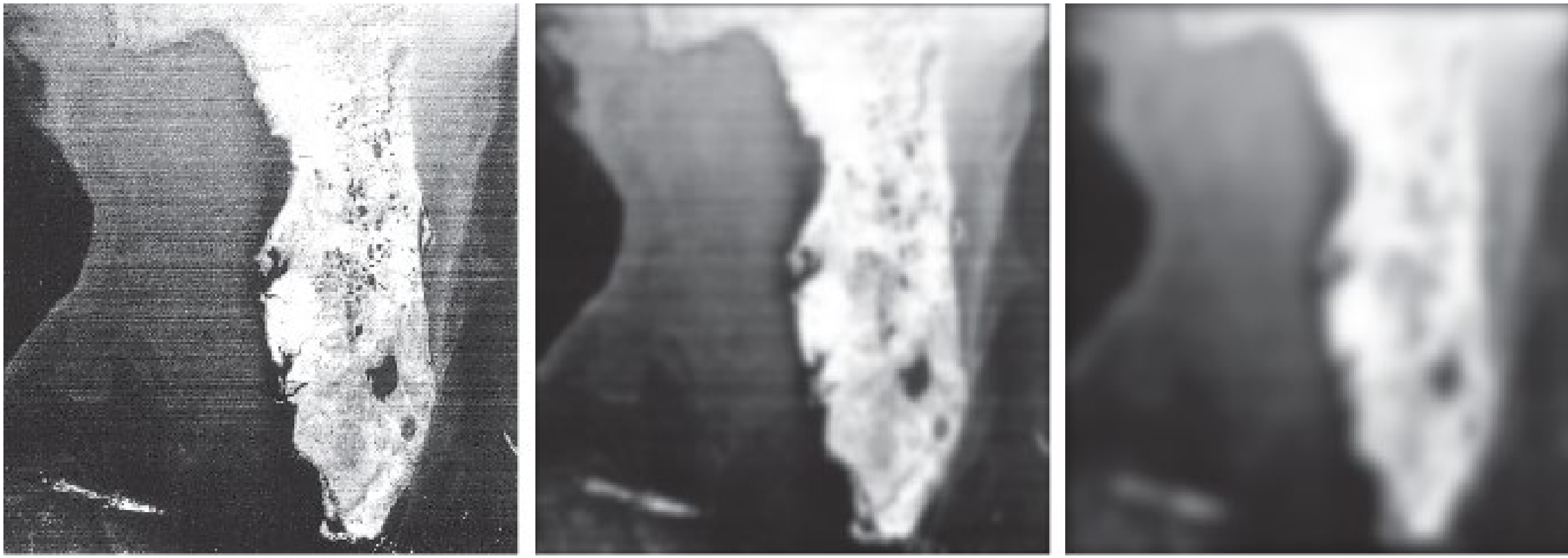
The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Example (GLPFs)

FIGURE 4.49 (a) Original 785×732 image. (b) Result of filtering using a GLPF with $D_0 = 150$. (c) Result of filtering using a GLPF with $D_0 = 130$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Example (GLPFs)



a b c

FIGURE 4.50 (a) 808×754 satellite image showing prominent horizontal scan lines. (b) Result of filtering using a GLPF with $D_0 = 50$. (c) Result of using a GLPF with $D_0 = 20$. (Original image courtesy of NOAA.)

Sharpening Frequency-Domain Filters

Concepts

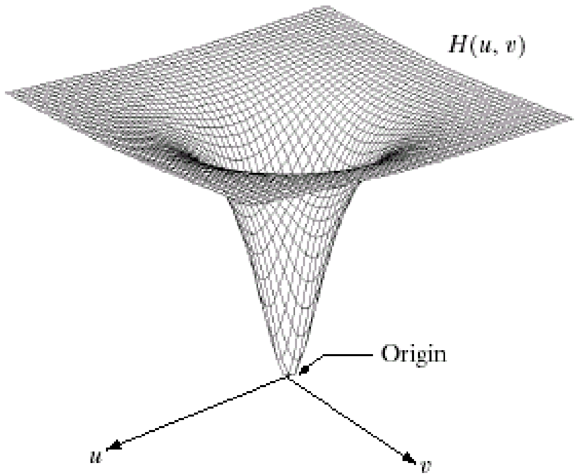
1. Because edges and other abrupt changes in grey levels are associated with high-frequency components, image sharpening can be achieved by using the high-pass frequency filtering process.

2. The high-pass filter attenuates (suppresses) the low-frequency components without disturbing high-frequency information in the frequency domain.

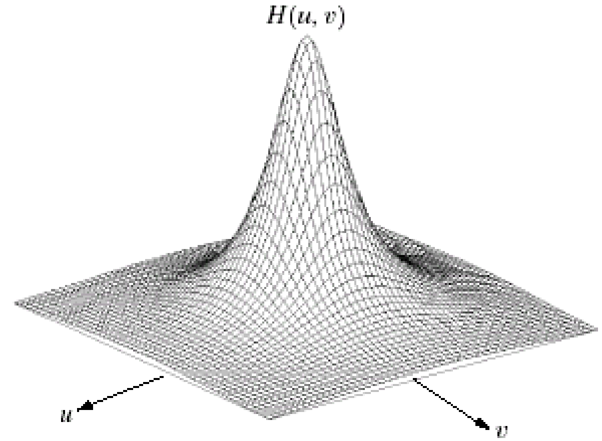
3. Relation between low-pass and high-pass filters

$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

(u,v) represents position variables in the frequency domain (frequency variables).



$= 1 -$



$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

Ideal High-Pass Filter (IHPF)

- Ideal Filter

  - $$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) <= D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

  - where $D(u,v)$ is the distance from point (u,v) to the origin of the frequency plane; $D(u,v) = (u^2 + v^2)^{1/2}$

  - D_0 is a non-negative quantity, cutoff frequency

Frequency representations

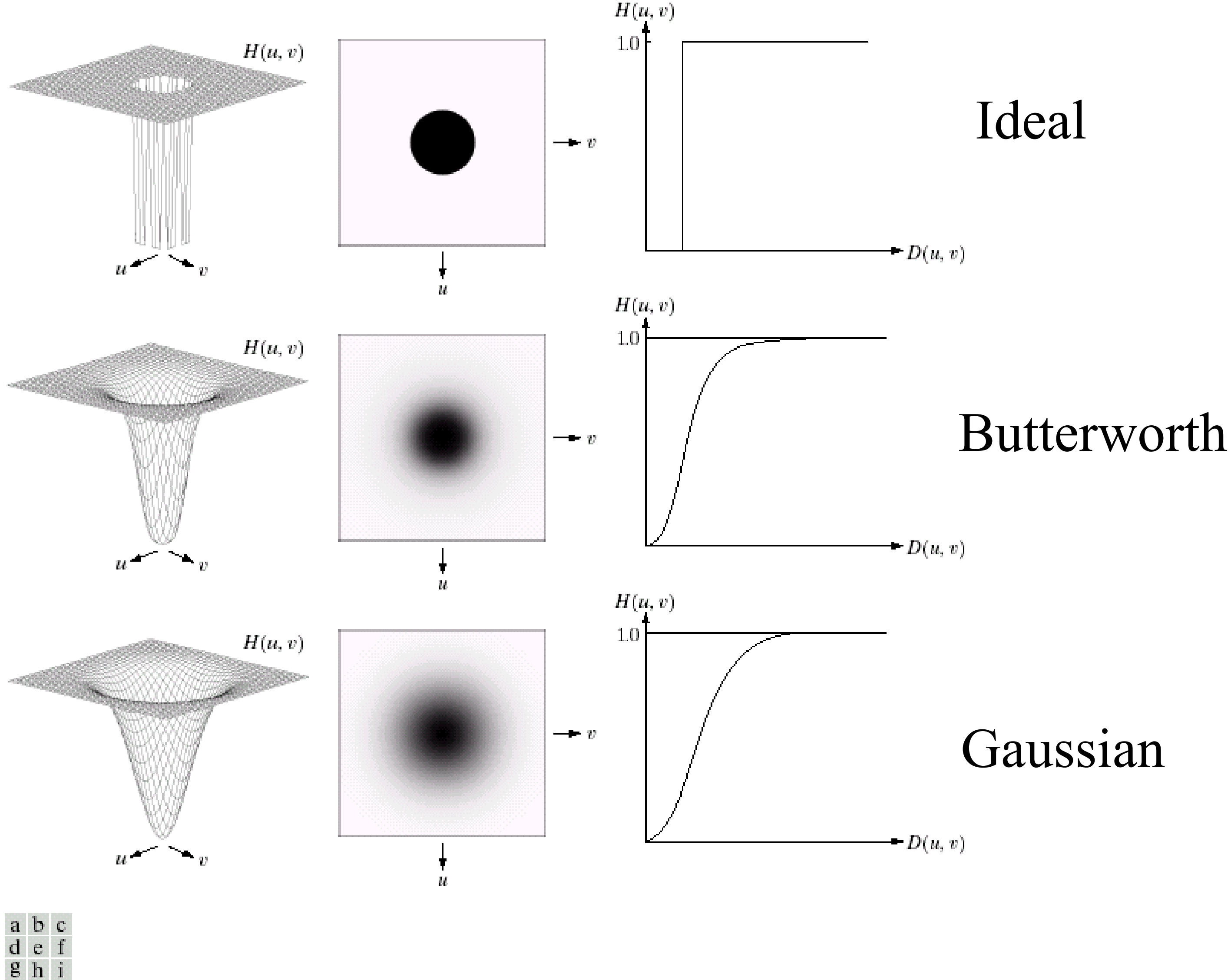
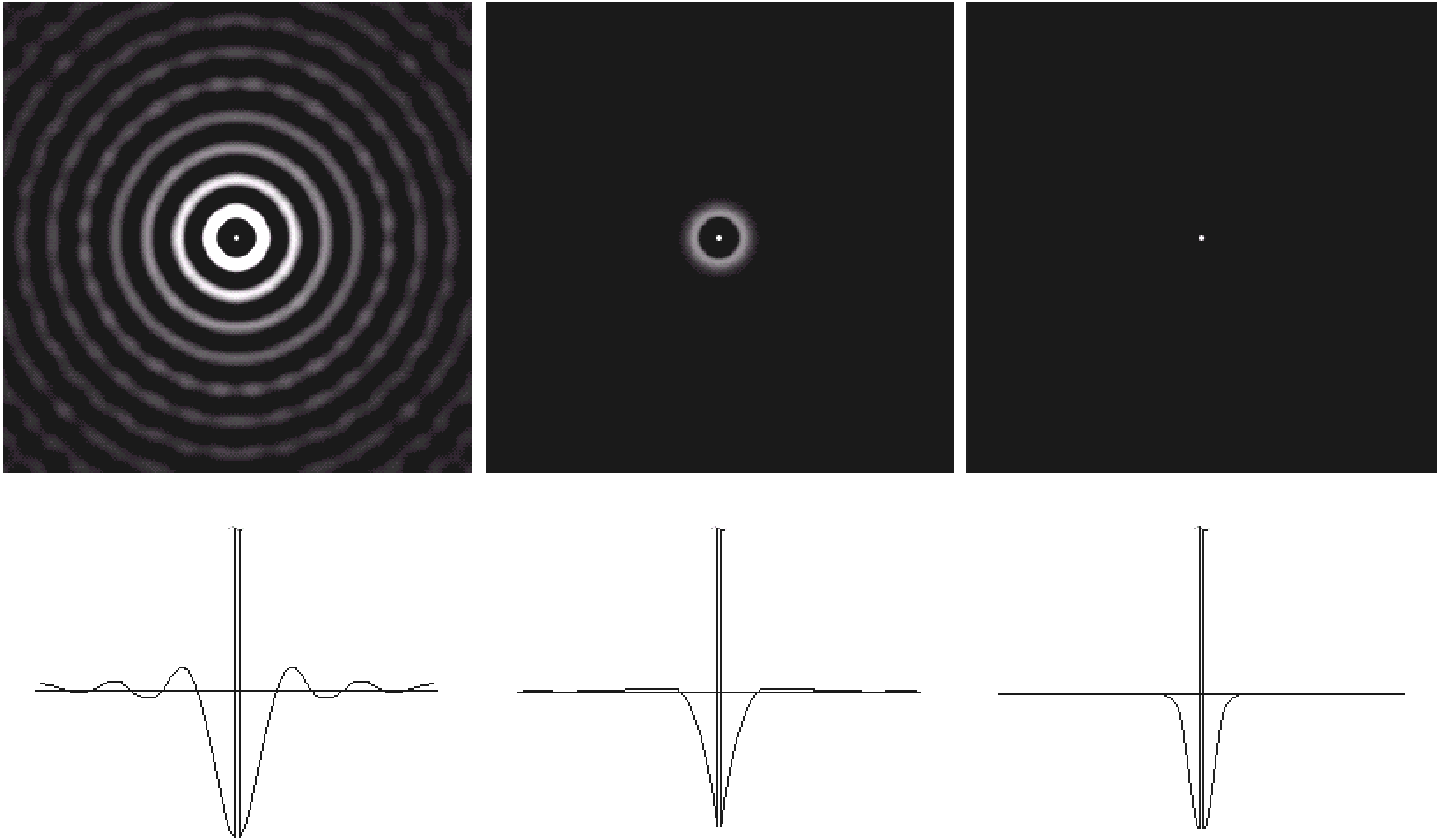


FIGURE 4.22 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Spatial representations



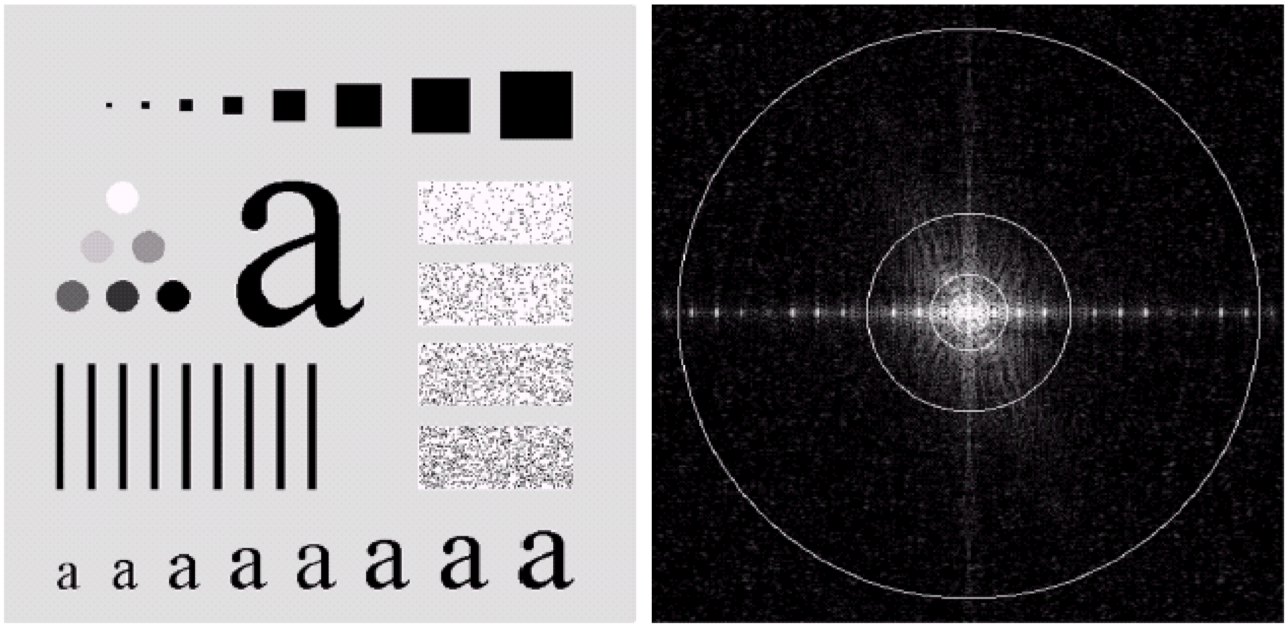
Horizontal intensity profiles

a b c

FIGURE 4.23 Spatial representations of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding gray-level profiles.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Input Image



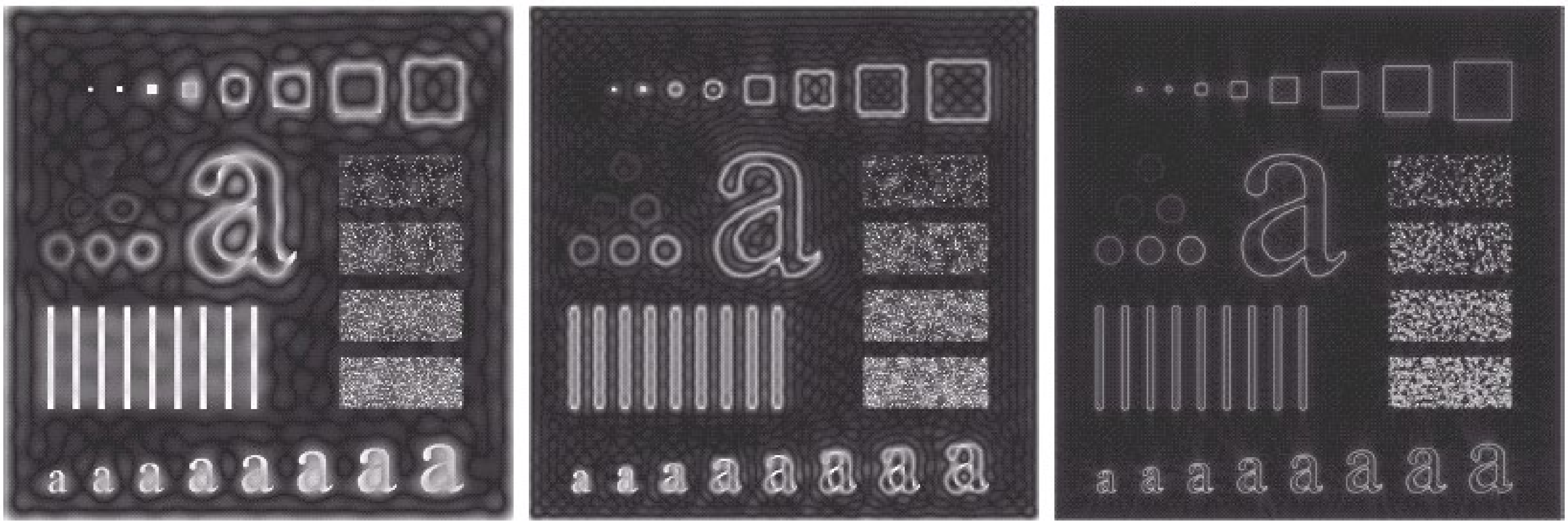
a b

FIGURE 4.11 (a) An image of size 500 × 500 pixels and (b) its Fourier spectrum. The superimposed circles have radii values of 5, 15, 30, 80, and 230, which enclose 92.0, 94.6, 96.4, 98.0, and 99.5% of the image power, respectively.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Ideal High-Pass Filter

- We can expect IHPFs to have the same ringing properties as ILPFs.



a b c

FIGURE 4.24 Results of ideal highpass filtering the image in Fig. 4.11(a) with $D_0 = 15$, 30, and 80, respectively. Problems with ringing are quite evident in (a) and (b).

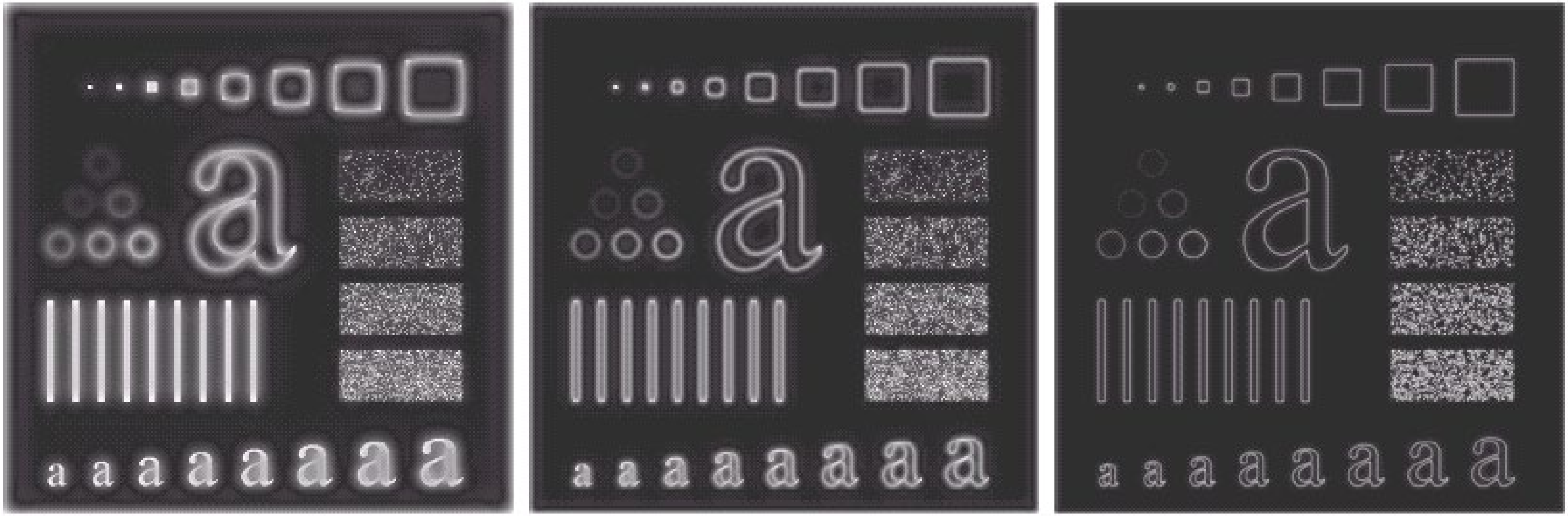
High-Pass Butterworth Filter

$$H(u,v) = \frac{1}{1+[\boxed{D_0 / D(u,v)}]^{2n}}$$

$$H(u,v) = \frac{1}{1+\alpha[D_0 / D(u,v)]^{2n}}$$

High-Pass Butterworth Filter

- We can expect Butterworth high-pass filters to behave more smoothly than IHPFs.
- The transition into higher values of cutoff frequencies is much smoother with the BHPF, as compared with the IHPF.



a b c

FIGURE 4.25 Results of highpass filtering the image in Fig. 4.11(a) using a BHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. These results are much smoother than those obtained with an ILPF.

Using a High-Pass Filter to Segment an Image

a b c

FIGURE 4.55 (a) Smudged thumbprint. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)

- Figure (a) is a 962 x 1026 image of a thumbprint with ridges (dim lines).
- Figure (b) is the result of using a Butterworth high-pass filter of order 4 with a cutoff frequency of 50.
- Figure (c) is the result of setting to black (0) all negative values and to white (1) the remaining values. Refer to Figure 4.23 of this set of notes.

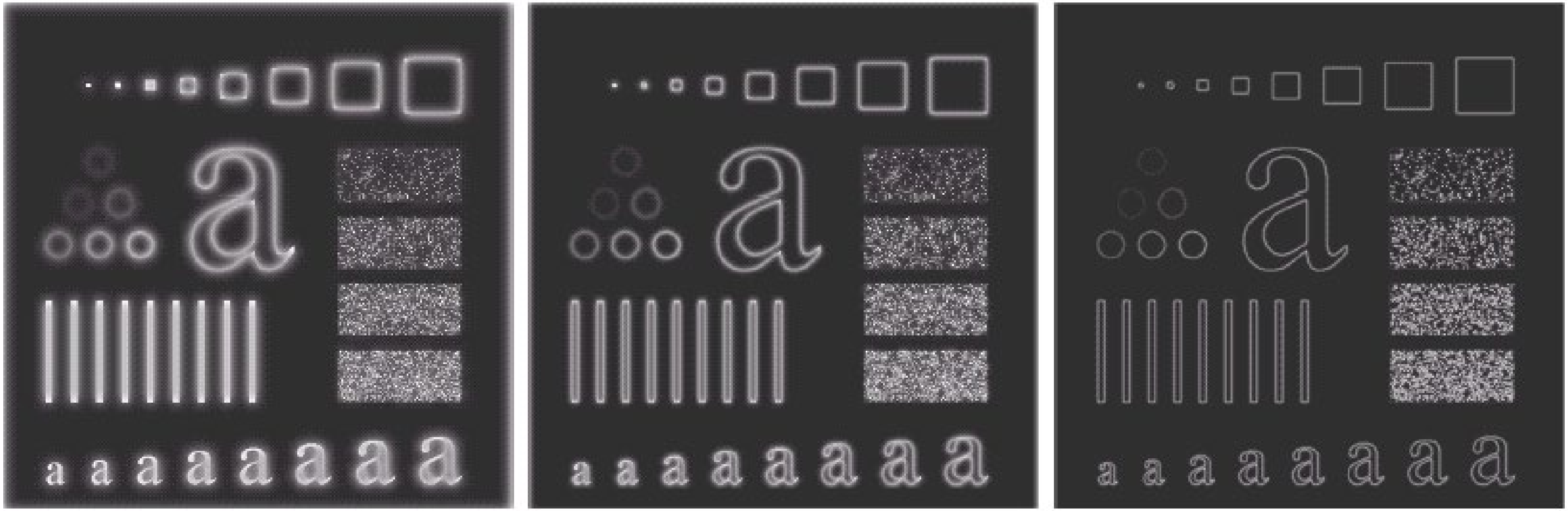
The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

High-Pass Gaussian Filter

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

- It is possible to construct high-pass Gaussian filters as the difference of low-pass Gaussian filters.
- Even the filtering of the smaller objects and thin bars is cleaner with the Gaussian filter.

High-Pass Gaussian Filter



a b c

FIGURE 4.26 Results of highpass filtering the image of Fig. 4.11(a) using a GHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. Compare with Figs. 4.24 and 4.25.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

The Laplacian in the frequency domain

a b

FIGURE 4.56 (a) Original, blurry image. (b) Image enhanced using the Laplacian in the frequency domain. Compare with Fig. 3.52(d). (Original image courtesy of NASA.)

$$g(x, y) = f(x, y) + c[\nabla^2 f(x, y)]$$

$G(u, v) =$ (given in class)

High-Frequency Emphasis

- Add a constant to the filter
- 0 <= a <= 1, b>a

$$H(u,v) = a + b\frac{1}{1+[D_0 / D(u,v)]^{2n}}$$

  - Preserves low-frequencies
  - Amplifies high-frequencies
  - (This technique is often used in conjunction with a histogram equalization)
  - Analogous to the “high-boost” filter

Examples

The result of GHPF, $D_0 = 70$

a b
c d

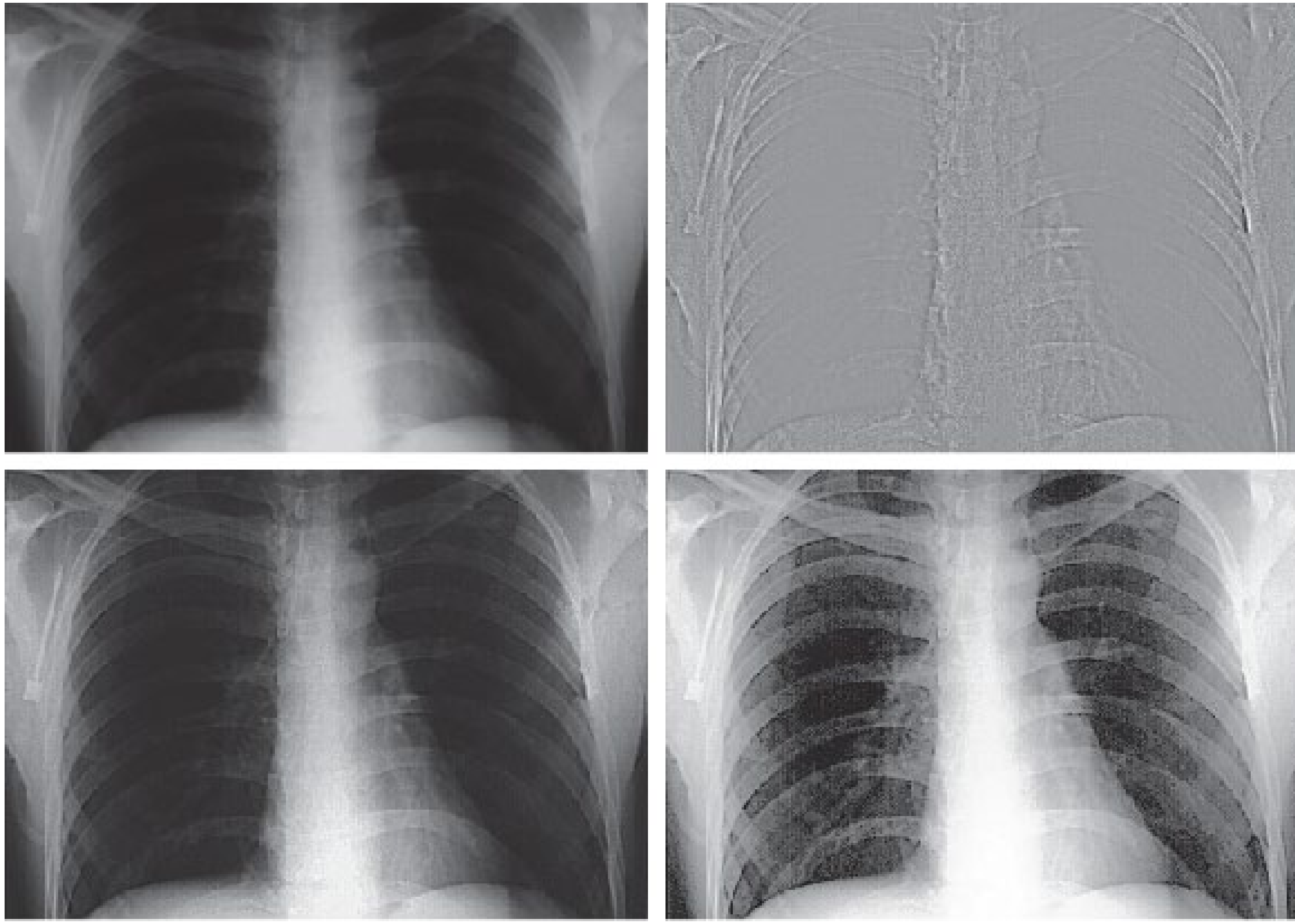


FIGURE 4.57
(a) A chest X-ray. (b) Result of filtering with a GHPF function. (c) Result of high-frequency-emphasis filtering using the same GHPF. (d) Result of performing histogram equalization on (c). (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

High-boost filtering,
$G(u,v) = [k_1 + k_2 H_{HP}(u,v)]F(u,v)$,
where $k_1 = 0.5$ and $k_2 = 0.75$

Histogram equalized result

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Homomorphic filtering

- Homomorphic filtering uses the illumination-reflectance model, $f(x,y) = i(x,y)r(x,y)$, where i = illumination and r = reflectance.
- The illumination and reflectance components of an image are separated by using log operations.
- The illumination component is represented by slow spatial variations, and the reflectance component is represented by the high frequencies, e.g., junctions of dissimilar objects.

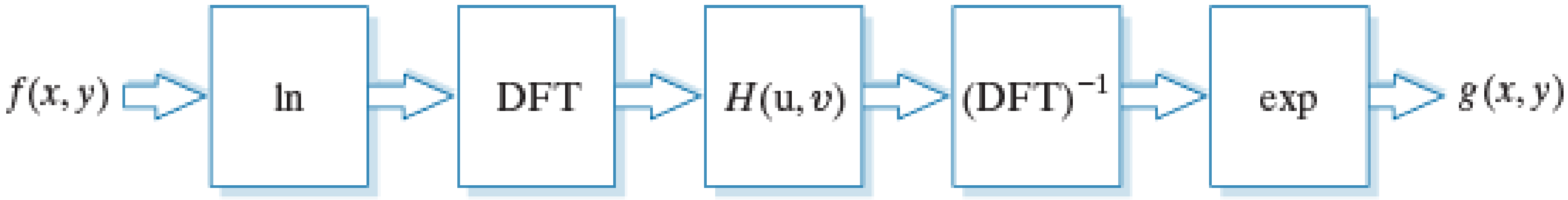


FIGURE 4.58 Summary of steps in homomorphic filtering.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Homomorphic filtering

- After these components are separated, a frequency domain filter is applied

$$H(u,v) = (\gamma_H - \gamma_L)\left[1 - exp\left(\frac{-CD^2(u,v)}{D_0^2}\right)\right] + \gamma_L$$

- The two parameters, γ_H, γ_L, controls the filtering of low- and high-frequency components.

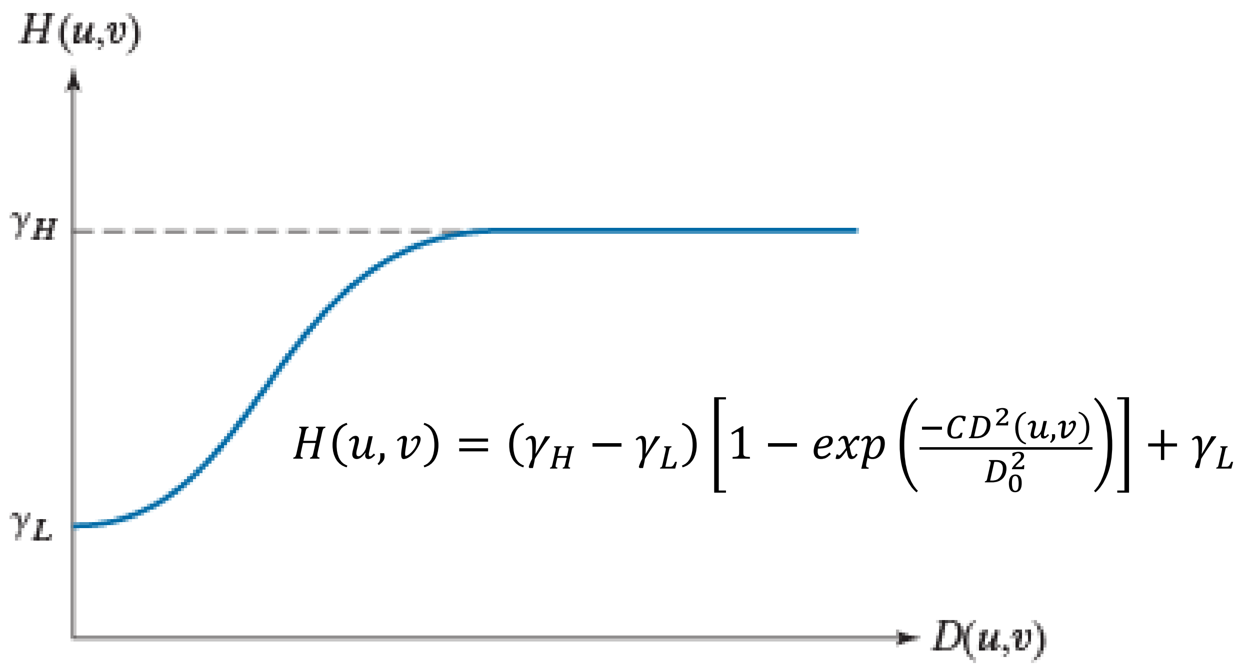
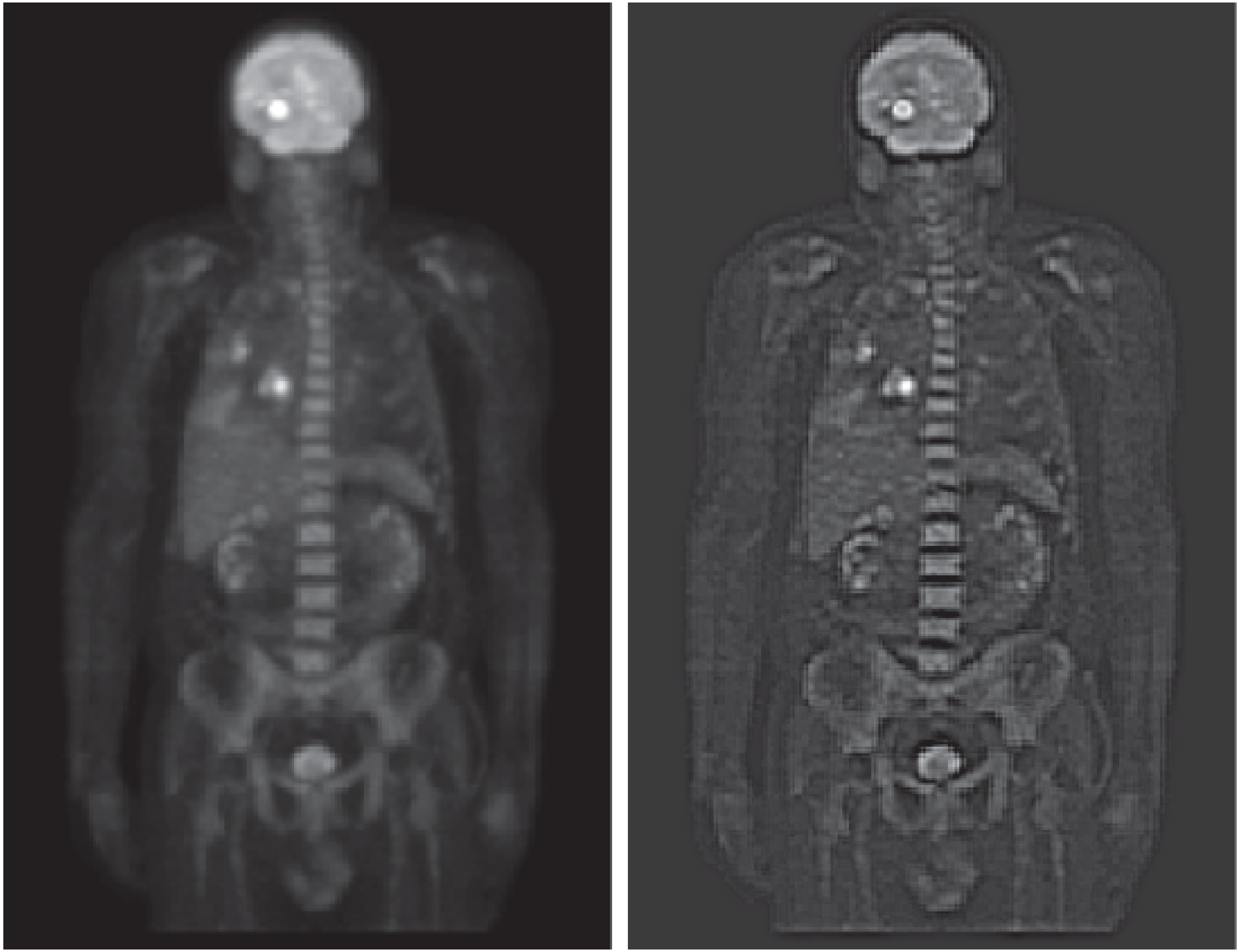


FIGURE 4.59 Radial cross section of a homomorphic filter transfer function.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Homomorphic filtering

$\gamma_L = 0.4$
$\gamma_H = 3.0$
$c = 5$
$D_0 = 20$



a b

FIGURE 4.60
(a) Full body PET scan. (b) Image enhanced using homomorphic filtering. (Original image courtesy of Dr. Michael E. Casey, CTI Pet Systems.)

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com